



PT SAPTAINDRA SEJATI

CYLINDER HEAD BEFORE MACHINING MEASUREMENT CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

HD785-7

RO NO :

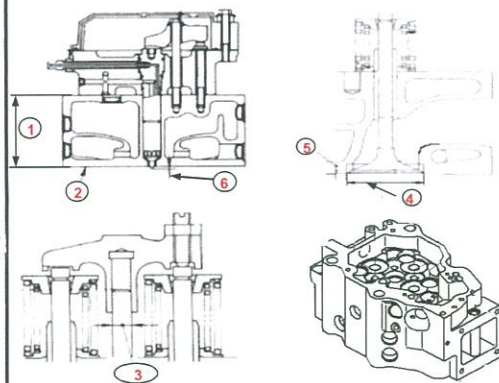
ENGINE SECTION

CYLINDER HEAD BEFORE MACHINING AND MEASUREMENT CHECK SHEET

WO NUMBER		PART NAME	CYLINDER HEAD	DOCUMENT NO.	RC-QA-EG-R2-0021
ENGINE MODEL	SAA12V140E-3	PART NUMBER		PUBL. DATE	
ENGINE S/N		START DATE / TIME		REVISION NO.	02
MACHINE MODEL		FINISH DATE / TIME		PAGE	

MEASURED BY		QUANTITY	4
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STANDARD FIGURES



No	Measuring Items			Std Tolerance (mm)	Repair Limit (mm)
1	Head Height			140.05 ~ 139.95	139.65
2	Lower Surface Distortion			0,060	0,090
3	O.D. of Cross Head Guide			11.011 ~ 11.000	10,950
4	Valve Seat Insert Bore	In	Std	55.000 ~ 55.019	-
			0.25 OS	55.250 ~ 55.269	-
			0.50 OS	55.500 ~ 55.519	-
		Ex	Std	52.000 ~ 52.019	-
			0.25 OS	52.250 ~ 52.269	-
			0.50 OS	52.500 ~ 52.519	-
5	Valve Sink			0 ± 0.10	0.70
6	Nozzle Protrusion			2.3 ~ 2.9	Replace nozzle

A.Result of Measurement

Result of Visual Check																
No.	Measuring Item			R1		R2		R3		R4		R5		R6		RANDOM INSPECTION
				MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	GL/SPV SIGN
1	Head Height															
2	Distortion															
3	O.D Cross Head Guide		In													
			Ex													
4	Valve Seat Insert Bore	In	1													
			2													
		Ex	1													
			2													
5	Valve Sink	In	1													
			2													
		Ex	1													
			2													
6	Nozzle protrution															
7	ID Valve Guide (Use Tool G/NG)	In	1	NEW PARTS												
			2													
		Ex	1													
			2													

CYLINDER HEAD BEFORE MACHINING AND MEASUREMENT CHECK SHEET

WO NUMBER		PART NAME	CYLINDER HEAD	DOCUMENT NO.	RC-QA-EG-R2-0021
ENGINE MODEL	SAA12V140E-3	PART NUMBER		PUBL. DATE	
ENGINE S/N		START DATE / TIME		REVISION NO.	02
MACHINE MODEL		FINISH DATE / TIME		PAGE	

[illegible]

B.Result of Measurement

[illegible]

CYLINDER HEAD BEFORE MACHINING AND MEASUREMENT CHECK SHEET

WO NUMBER		PART NAME	CYLINDER HEAD	DOCUMENT NO.	RC-QA-EG-R2-0021
ENGINE MODEL	SAA12V140E-3	PART NUMBER		PUBL. DATE	
ENGINE S/N		START DATE / TIME		REVISION NO.	02
MACHINE MODEL		FINISH DATE / TIME		PAGE	

Unit : mm

3 of 4

ENGINE MODEL	VALVE SPRING	STANDARD SIZE		REPAIR LIMIT
		FREE LENGTH	INSTAL LENGTH	LOAD (KG)
12V140 Series	INNER	72,00	46,00	19,70
	OUTER	81,70	51,00	38,70
6D140 Series	INNER	72,00	46,00	19,60
	OUTER	81,70	51,00	38,50
6D170-3	INNER	-	-	-
	OUTER	99,60	87,40	62,00
6D170 Series	INNER	59,30	47,00	17,40
	OUTER	64,80	37,50	84,80
6D170 HD465-5	INNER	73,80	48,00	23,30
	OUTER	77,00	52,00	48,70
6D125-2	INNER	-	-	-
	OUTER	75,20	56,00	47,70
6D155-4	INNER	-	-	-
	OUTER	88,00	51,90	65,50

[illegible]

CYLINDER HEAD BEFORE MACHINING AND MEASUREMENT CHECK SHEET

WO NUMBER		PART NAME	CYLINDER HEAD	DOCUMENT NO.	RC-QA-EG-R2-0021
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ENGINE S/N		START DATE / TIME		REVISION NO.	02
MACHINE MODEL		FINISH DATE / TIME		PAGE	

Measurement procedure of flatness Cylinder head surface

Method of measurement of depth, and measurement tool :
Visual inspection or use Dial gauge.

Judgement of Distortion(Flatness)

Measuring point : Refer the right picture



Upper surface	Flatness	Remedy
	within 0.08mm	Use as it is
	More than 0.08mm	Reuse after surface grinding
lower surface	within 0.08mm	Reuse after surface grinding

Measurement of Distortion , and measurement tool
Straight edge, Clearance gauge

Measure flatness of the bottom surface of head.

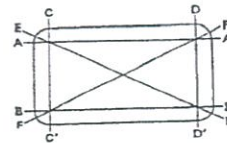


Fig. A



Fig. B



INSPECTION RESULT :

[illegible]

INSPECTION BY :

INSPECTION & APPROVAL BY :

MECHANIC (MEC)

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION



PT SAPTAINDRA SEJATI

**CYLINDER HEAD AFTER
MACHINING MEASUREMENT
CHECKSHEET**

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

HD785-7

RO NO :

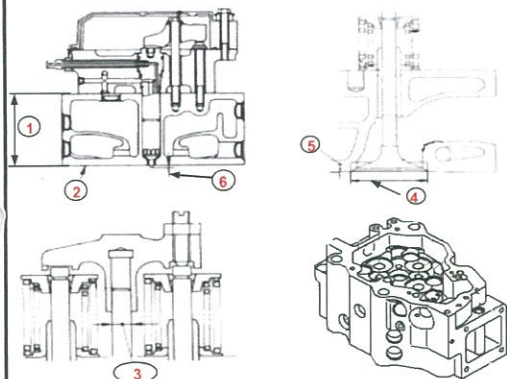
ENGINE SECTION

CYLINDER HEAD AFTER MACHINING AND MEASUREMENT CHECK SHEET

WO NUMBER		PART NAME	CYLINDER HEAD	DOCUMENT NO.	RC-QA-EG-R2-0021
ENGINE MODEL	SAA12V140E-3	PART NUMBER		PUBL. DATE	
ENGINE S/N		START DATE / TIME		REVISION NO.	02
MACHINE MODEL		FINISH DATE / TIME		PAGE	

MEASURED BY		QUANTITY	4
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STANDARD FIGURES



No	Measuring items			Std Tolerance (mm)	Repair Limit (mm)
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A.Result of Measurement

A.Result of Measurement																	
No.	Measuring Item			R1		R2		R3		R4		R5		R6		RANDOM INSPECTION	
				MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	GL/SPV SIGN	
1	Head Height																
2	Distortion																
3	O.D Cross Head Guide	In															
		Ex															
4	Valve Seat Insert Bore	In	1	MEASURED BY DIS-ASSEMBLY SECTION													
			2														
		Ex	1														
			2														
5	Valve Sink	In	1														
			2														
		Ex	1														
			2														
6	Nozzle protrution																
7	ID Valve Guide (Use Tool G/NG)	In	1	NEW PARTS													
			2														
		Ex	1														
			2														

MEASURED BY DIS-ASSEMBLY SECTION

NEW PARTS

CYLINDER HEAD AFTER MACHINING AND MEASUREMENT CHECK SHEET

[illegible]



PT SAPTAINDRA SEJATI

CYLINDER HEAD MACHINING MEASUREMENT CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

HD785-7

RO NO :

ENGINE SECTION

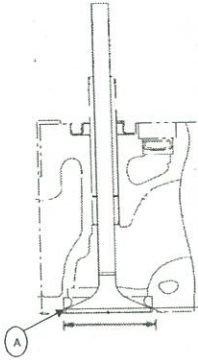


PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

CYLINDER HEAD MACHINING AND MEASUREMENT CHECKSHEET

JOB NUMBER		PART NAME	CYLINDER HEAD	DOCUMENT NO.	RC-QA-EG-R2-0023
ENGINE MODEL	ALL SERIES	PART NUMBER		PUBL. DATE	
ENGINE S/N		START DATE/TIME		REVISION NO.	02
MACHINE MODEL		FINISH DATE/TIME		PAGE	1 of 1

SKETCH DRAWING



Standard size

ENGINE MODEL	VALVE SINK WITHOUT SPRING (A)		VACUM TEST (BAR)
	STANDARD(mm)	REPAIR LIMIT(mm)	
12V140	0 ± 0,10	0,70	- (6 ~ 8)
6D170-5	0 ± 0,10	0,80	
6D170E-3	0 ± 0,10	0,40	
6D140E-1	0 ± 0,10	0,70	
6D125-1	0 ± 0,10	0,70	

MAN HOURS UTILIZATION			
PROCESS	START	FINISH	MAN HOURS
Salvaging			
Measuring			
TOTAL			

VISUAL CHECK	MECHANIC		QA	
	OK	NOT OK	OK	NOT OK
CRACK HAS BEEN CHECK				

No.	Measuring Item			R1		R2		R3		R4		R5		R6		RANDOM INSPECTION
				MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	
1	Valve Sink	In	1													
			2													
		Ex	1													
			2													
2	Vacuum Test	STD (6~8 Bar)	IN													
			EX													
	Measuring Item			L1		L2		L3		L4		L5		L6		RANDOM INSPECTION
				MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	MEC	QA/GL	
3	Valve Sink	In	1													
			2													
		Ex	1													
			2													
4	Vacuum Test	STD (6~8 Bar)	IN													
			EX													

INSPECTION RESULT :

DELIVERY BY MACHINING	
DELIVERY DATE :	
MACHINIST NAME :	
SIGN	

RECEIVING INSPECTION BY SUB ASSY	
RECEIVED DATE :	
MECHANIC NAME :	
SIGN	

Note : New Cyl Head & after finishing can be directly approval by QA (Quality Assurance)



PT SAPTAINDRA SEJATI

CYLINDER HEAD ASSEMBLY CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

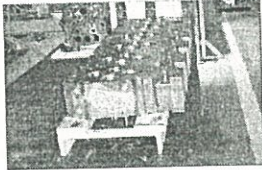
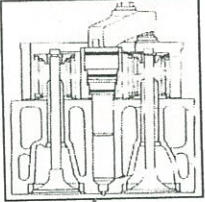
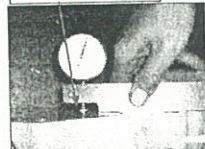
HD785-7

RO NO :

ENGINE SECTION

ASSEMBLY CHECK SHEET CYLINDER HEAD ENGINE ALL SERIES	WO NO.		PART NAME	CYL. HEAD ASS'Y	DOCUMENT NO.	
	ENGINE MODEL		PART NUMBER		PUBLIC DATE	13/10/2009
	ENGINE S/N		START DATE / TIME		REVISION NO.	01
	MACHINE MODEL		FINISH DATE / TIME		PAGE	2 of 2

ASSEMBLY BY	All dimensions in millimetres
	All torques in Kilogram metres

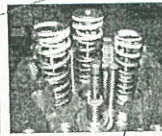
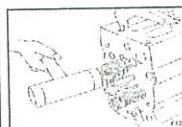
NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN			REMARKS (SKETCH, PHOTOS & COMMENT)
					MEC	QA	SPV	
1	CYL. HEAD ASSEMBLY	Head Assy Threads Cyl. Heads	- Head Assy must be free from crack, dents, pitting, rust and worn - Make sure all threads are not damage - Make sure top surface and bottom surface of the Cylinder heads are good conditions - Make sure the Cylinder Heads are clean					
2	VALVE GUIDES	Bores	- The Valve Guides be renew after measured. - Make sure the Valve Guide bores are clean when reinstall of the Valve Guides - Before inserting the valve guide, coat the outer dimension of the valve guide with small quantity of engine oil (SAE 15W-40)	GO and NO GO Hyd. Press				 Insert and press valve guides
3	INSERT VALVE	Insert Valve Bores Valve Sinking	- Make sure the dimension of the Insert Valve bores - Check that there is no dust, crack, corrosive and dirt - Install the Insert Valve be carefully - Diameter between Insert Valve bore and Insert Valve must interference see measurement check sheet - When install Insert Valves must straight to hole position - Check Valve sinking/Protrusion and refer to Standard (Shop Manual or measurement Check Sheet) - If sinking or Protrusion out of standard, cutting Insert or Valve and recheck sinking or Protrusion it must standard condition.	Special Tool Grinding Valve machine or Cutting Insert machine				 Install insert valve  Check valve sinking 
4	NOZZLE SLEEVE	Nozzle Sleeve Condition	- The Nozzle Sleeves must be renew - Check the O-ring grooves and the Nozzle Sleeve bores are clean when reinstall of the Nozzle Sleeves - Make sure the Nozzle Sleeves are in good position - When install and cutting the Nozzle Sleeves must use the Special Tool and refer to replacing Nozzle Holder Sleeve in Shop Manual (see replacing nozzle holder sleeve Job Instruction). - Check and measure protrusions of Nozzles after cutting Nozzle Sleeves (See and Record at the Measurement Check Sheet).					 Check nozzle protrusions 
5	CRACK & LEAK TEST CYL. HEAD	Pressure test	- Air Pressure Test - Assemble plate and flange - Connect a hose to flange - Place the Cyl. Head in a water bath with hot water - Apply air pressure (3.0 ~ 3.5 kg/cm²) for approx 5 ~ 10 Min. - Check for any air leakage in the water. - Replace Cyl. Head if there are crack.	3.0 ~ 3.5 kg/cm ² for approx 5 ~ 10 Min. (82°C ~ 93°C)				 Check cylinder head with hot water & air pressure

ASSEMBLY CHECK SHEET

CYLINDER HEAD

ENGINE ALL SERIES

WO NO.		PART NAME	CYL. HEAD ASS'Y	DOCUMENT NO.	
ENGINE MODEL		PART NUMBER		PUBLIC DATE	13/10/2009
ENGINE S/N		START DATE /TIME		REVISION NO.	01
MACHINE MODEL		FINISH DATE / TIME		PAGE	2 of 2

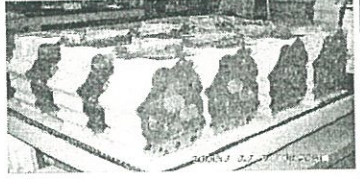
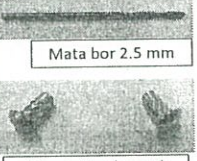
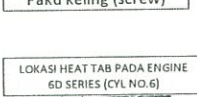
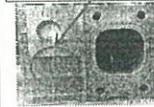
NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN			REMARKS (SKETCH, PHOTOES & COMMENT)																					
					MEC	QA	SPV																						
6	VALVE & SPRING ASSEMBLY		- Make sure Cylinder Head free from Dirt and foreign material - Make sure all component that will assembly already clean 1. Align with casting mark on Cylinder Head and assembly Intake valve and Exhaust valve coat the valve steam and the face of the Valve thinly with engine oil (SAE 15W ~ 40) ● Make sure of Valve (Intake and Exhaust) correct arrangement ● Make sure the Valves are following to PSN Valve code				 Install lower seat  Install valve seal  Install outer spring  Install inner spring  Compress spring  Fit valve cotter  																						
		PSN	<table><tr><th>ENGINE MODEL</th><th>VALVE INTAKE</th><th>VALVE EXHAUST</th></tr><tr><td>140 SERIES</td><td>I</td><td>CX / PX</td></tr><tr><td>170 SERIES</td><td>AN / BN</td><td>EX / X</td></tr><tr><td>125 SERIES</td><td>IN / BN</td><td>EX / BX</td></tr><tr><td>155 SERIES</td><td>I</td><td>E</td></tr><tr><td>170-3</td><td>IN</td><td>E</td></tr><tr><td colspan="3">ACTUAL</td></tr><tr><td></td><td></td><td></td></tr></table>	ENGINE MODEL	VALVE INTAKE	VALVE EXHAUST	140 SERIES	I	CX / PX	170 SERIES	AN / BN	EX / X	125 SERIES	IN / BN	EX / BX	155 SERIES	I	E	170-3	IN	E	ACTUAL							
ENGINE MODEL	VALVE INTAKE	VALVE EXHAUST																											
140 SERIES	I	CX / PX																											
170 SERIES	AN / BN	EX / X																											
125 SERIES	IN / BN	EX / BX																											
155 SERIES	I	E																											
170-3	IN	E																											
ACTUAL																													

INSPECTION RESULT :



PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

ASSEMBLY CHECK SHEET CYLINDER HEAD ENGINE ALL SERIES	WO NO.		PART NAME	CYL. HEAD ASS'Y	DOCUMENT NO.	
	ENGINE MODEL		PART NUMBER		PUBLIC DATE	13/10/2009
	ENGINE S/N		START DATE / TIME		REVISION NO.	01
	MACHINE MODEL		FINISH DATE / TIME		PAGE	2 of 2

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN			REMARKS (SKETCH, PHOTOS & COMMENT)
					MEC	QA	SPV	
7	HEAT TAB	PREPARE CYL HEAD	- Prepare cylinder head yang akan dipasang heat tab - Letakkan cylinder head ditempat yang rata <i>NOTE: Untuk pemasangan heat tab tidak dipasang disetiap cylinder head. Berikut referensi nya:</i> a. engine 6D series hanya 1 cylinder head (no.6) b. engine 12V series hanya 2 cylinder head (no. 6 RH dan no. 6 LH)					
		TOOLS	- Siapkan tools untuk membuat lubang paku rivet pada cyl head - Siapkan Mesin bor (stand) - Siapkan mata bor 2.5 mm - Siapkan paku keling (screw) - Siapkan Heat tab baru (new)					 Mata bor 2.5 mm  Mesin bor  Paku keling (screw)  Heat tab baru, posisi 3 lubang pada heat tab LOKASI HEAT TAB PADA ENGINE 6D SERIES (CYL NO.6)
		ASSEMBLY	- Letakkan cyl head ditempat yang rata - Bor diarea exhaust untuk engine 6D series menggunakan mata bor 2.5 mm. - Posisi lubang untuk paku keling 3 titik, sesuai dengan lubang yang ada pada heat tab. - Kedalaman bor pada cyl head <u>Actual :</u> - Setelah selesai mengebor, install heat tab dan paku keling	STD : 3.0 ~ 3.5 mm				LOKASI HEAT TAB UNTUK ENGINE 12V SERIES LH BANK (CYL NO.6)  LOKASI HEAT TAB PADA ENGINE 12V SERIES RH BANK (CYL NO.6)  Heat tab ready install 

INSPECTION RESULT :

ASSEMBLY BY :

INSPECTION & APPROVAL BY :

MECHANIC (MEC)

QUALITY ASSURANCE (QA)

PROD. SUPERVISOR (SPV)



PT SAPTAINDRA SEJATI

TIMING GEAR

MEASUREMENT & INSPECTION

CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

HD785-7

RO NO :

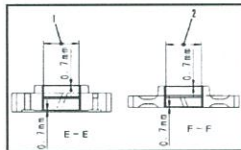
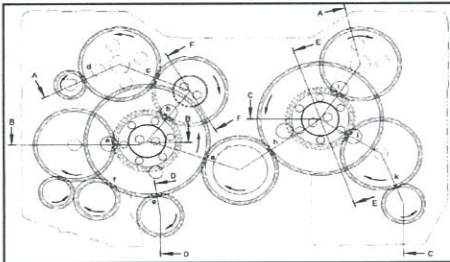
ENGINE SECTION



SALVAGE IDLER GEAR AND SHAFT PROCESS INSPECTION CHECKSHEET

WO NUMBER		PART NAME	IDLER GEAR AND SHAFT	DOCUMENT NO.	RC/QR04/EG/10/01
ENGINE MODEL	12V140 -3	PART NUMBER		PUBL. DATE	
ENGINE S/N		START DATE/TIME		REVISION NO.	00
MACHINE MODEL		FINISH DATE/TIME		PAGE	1 of 1

SKETCH DRAWING



Physical Data

No.	Part Number	Part Name	Qty	New	Reuse
1					
2					
3					
4					

MACHINIST NAME

Standard size (unit mm)

DESCRIPTION	STANDARD DIMENSION		CLEARANCE	
	SHAFT	BUSHING	STANDARD	REPAIR LIMIT
1. Main Idler Gear (R and L Bank)	55.984 ~ 55.971	56.034 ~ 56.009	0.025 ~ 0.063	0,20
2. Sub Idler Gear	55.984 ~ 55.971	56.055 ~ 56.000	0.016 ~ 0.084	0,20

VISUAL CHECK

MECHANIC

QA/GL

OK

NOT OK

OK

NOT OK

CRACK HAS BEEN CHECKED

MANHOURS UTILIZATION

PROCESS	START	FINISH	MAN HOURS
Machining			
Measuring			
TOTAL			

INSPECTION SALVAGE PROCESS

SKETCH DRAWING	DIMENSION		ACTUAL DIAMETER						ACTUAL CLEARANCE			ROTATING CHECK (OK or NOT)		
			SHAFT			BUSHING			MAC	QA/GL	SPV	MAC	QA/GL	SPV
			MAC	QA/GL	SPV	MAC	QA/GL	SPV						
SUB IDLER GEAR 	A	X							MIN					
		Y												
	B	X												
		Y								MAX				
	C	X												
		Y												
MAIN IDLER GEAR LH 	A	X							MIN					
		Y												
	B	X												
		Y								MAX				
	C	X												
		Y												
MAIN IDLER GEAR RH 	A	X							MIN					
		Y												
	B	X												
		Y								MAX				
	C	X												
		Y												

MECHANIC INSPECTION COMMENT :

INSPECTION BY :

INSPECTION & APPROVED BY :

MACHINIST (MACH)

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION



PT SAPTAINDRA SEJATI

ROCKER ARM & SHAFT MEASUREMENT & POLISHING CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

HD785-7

RO NO :

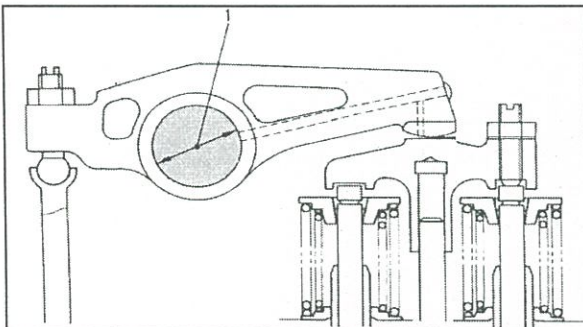
ENGINE SECTION



ROCKER ARM SHAFT MEASURING AND POLISHING CHECK SHEET

WO NUMBER		PART NAME	ROCKER ARM SHAFT	DOCUMENT NO.	QR08-SAL-003-07
ENGINE MODEL	SA12V140-1/3	PART NUMBER		PUBL. DATE	05/02/2008
ENGINE S/N		START DATE/TIME		REVISION NO.	00
MACHINE MODEL		FINISH DATE/TIME		PAGE	1 of 1

SKETCH DRAWING



MACHINIST NAME

MAN HOURS UTILIZATION

PROCESS	START	FINISH	MAN HOURS
Machining			
Measuring			
TOTAL			

(Unit : mm)

ENGINE MODEL	STANDARD	STANDARD TOLERANCE	CLEARANCE LIMIT
12V140-1,-2,-3	Ø 32.0000	Ø 32.0065 ~ 31.9935	0,13

INSPECTION MEASURING PROCESS

CYL.NO.	CHECK POSITION	1			2			3			Actual Clearance
		MAC	QA	SPV	MAC	QA	SPV	MAC	QA	SPV	
RH	1	X-X'									
		Y-Y'									
	2	X-X'									
		Y-Y'									
	3	X-X'									
		Y-Y'									
LH	1	X-X'									
		Y-Y'									
	2	X-X'									
		Y-Y'									
	3	X-X'									
		Y-Y'									

MACHINIST INSPECTION RESULT :

INSPECTION BY :

INSPECTION & APPROVED BY :

MACHINIST (MACH)

QUALITY ASSURANCE (QA)

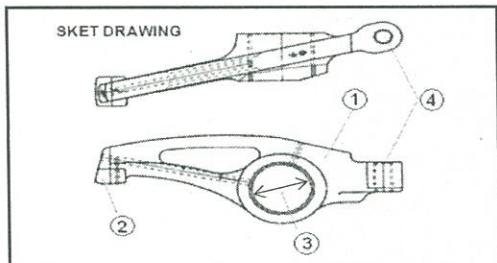
PROD SUPERVISOR (SDV)



PT. SAPTA INDRA SEJATI
REBUILD CENTER

ROCKER ARM POLISHING AND INSPECTION CHECK SHEET

JOB NUMBER		PART NAME	ROCKER ARM	DOCUMENT NO.	
ENGINE MODEL	SAA12V140-3	PART NUMBER		PUBL. DATE	
ENGINE S/N		START DATE/TIME		REVISION NO.	00
MACHINE MODEL		FINISH DATE/TIME		PAGE	1 of 1



MACHINIST NAME	
----------------	--

MAN HOURS UTILIZATION			
PROCESS	START	FINISH	MAN HOURS
Polishing			
Inspection			
TOTAL			

ROCKER ARM (LH)					√ = Good						X = No Good						
No	Check Point	Check Content	Standard Reuseble Parts	Methode	L1		L2		L3		L4		L5		L6		Remarks
					IN	EX	IN	EX	IN	EX	IN	EX	IN	EX	IN	EX	
1	Arm assy	Scrathes	No scratch	Color check													
		Appearance	Rivet no come out	Visual check													
		Oil hole	No clog	Visual check													
2	Contact surface	Scrathes	No scratch	Visual check													
		Crack	No crack	Color check													
		Wear	No stepped type wear	Visual check													
3	Bush inside	Scratches	No scratch	Visual check													
		Wear	Max. 32.098 mm	Measurement													
4	Screw Portion	Wear	No wear	Visual check													
FINAL INSPECTION BY QA																	
FINAL INSPECTION BY SUPERVISOR																	

INSPECTION OF ROCKER ARM					√ = Good								X = No Good					
No	Check Point	Check Content	Standard Reuseble Parts	Methode	R1		R2		R3		R4		R5		L6		Remarks	
					IN	EX	IN	EX	IN	EX	IN	EX	IN	EX	IN	EX		
1	Arm assy	Scrathes	No scratch	Visual check														
		Appearance	Rivet no come out	Visual check														
		Oil hole	No clog	Visual check														
2	Contact surface	Scrathes	No scratch	Visual check														
		Crack	No crack	Visual check														
		Wear	No stepped type wear	Visual check														
3	Bush inside	Scratches	No scratch	Visual check														
		Wear	Max. 32.098 mm	Measurement														
4	Screw Portion	Wear	No wear	Visual check														
FINAL INSPECTION BY QA																		
FINAL INSPECTION BY GL																		

INSPECTION COMMENT :	
INSPECTION BY :	INSPECTION & APPROVED BY :
MACHINIST	QUALITY ASSURANCE (QA)
	GL/SPV PRODUCTION



PT SAPTAINDRA SEJATI

CAM FOLLOWER & PUSHROD MEASUREMENT CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

HD785-7

RO NO :

ENGINE SECTION

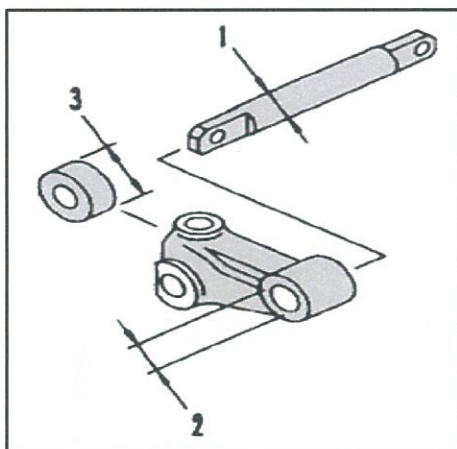
CAM FOLLOWER AND PUSH ROD MEASUREMENT CHECK SHEET

WO NUMBER		PART NAME	CAM FOLLOWER & PUSHROD	DOCUMENT NO.	RC/QR03/EG/10/01
ENGINE MODEL	12V140-1,3 Series	PART NUMBER		PUBL. DATE	10/08/2008
ENGINE S/N		START DATE/TIME		REVISION NO.	04
MACHINE MODEL		FINISH DATE/TIME		PAGE	

1 of 5

MEASURED BY	
-------------	--

STANDARD FIGURE



Unit : mm

ENGINE MODEL	CAM FOLLOWER RADIAL PLAY	
	STANDARD	Repair Limit
12V140-1,2 series	0.077 ~ 0.0150	0,151

Unit : mm

PUSH ROD BENDING STANDARD	
TIR max.	0.30

12V140-1

Unit : mm

NO	CHECK ITEM	DIMENSION		
		SIZE	STD TOLERANCE	Repair Limit
1	OD of C.F. Shaft	22	21.979 ~ 21.966	21,92
2	ID of C.F. Shaft Hole	22	22.000 ~ 22.021	22,07
3	OD Cam Roller	31,7	31.750 ~ 31.720	31,5



RADIAL PLAY CAM FOLLOWER						
CAM FOLLOWER RH	IN			EX		
	MECHANIC	QA	SPV	MECHANIC	QA	SPV
1						
2						
3						
4						
5						
6						

CAM FOLLOWER LH	IN			EX		
	MECHANIC	QA	SPV	MECHANIC	QA	SPV
1						
2						
3						
4						
5						
6						

CAM FOLLOWER AND PUSH ROD MEASUREMENT CHECK SHEET

WO NUMBER		PART NAME	CAM FOLLOWER & PUSH ROD	DOCUMENT NO.	RC/QR03/EG/10/01
ENGINE MODEL	12V140-1,3 Series	PART NUMBER		PUBL. DATE	10/08/2008
ENGINE S/N		START DATE/TIME		REVISION NO.	04
MACHINE MODEL		FINISH DATE/TIME		PAGE	

2 of 5

1	OUTSIDE DIAMETER OF CAM FOLLOWER SHAFT					
OD of CF Shaft RH	X			Y		
	MECHANIC	QA	SPV	MECHANIC	QA	SPV
1						
2						
3						
4						
5						
6						

1	OUTSIDE DIAMETER OF CAM FOLLOWER SHAFT					
OD of CF Shaft LH	X			Y		
	MECHANIC	QA	SPV	MECHANIC	QA	SPV
1						
2						
3						
4						
5						
6						

2	INSIDE DIAMETER OF CAM FOLLOWER HOLE					
ID of CF Shaft Hole RH	X			Y		
	MECHANIC	QA	SPV	MECHANIC	QA	SPV
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						

CAM FOLLOWER AND PUSH ROD MEASUREMENT CHECK SHEET

WO NUMBER		PART NAME	CAM FOLLOWER & PUSHROD	DOCUMENT NO.	RC/QR03/EG/10/01
ENGINE MODEL	12V140-1,3 Series	PART NUMBER		PUBL. DATE	10/08/2008
ENGINE S/N		START DATE/TIME		REVISION NO.	04
MACHINE MODEL		FINISH DATE/TIME		PAGE	

3 of 5

2	INSIDE DIAMETER OF CAM FOLLOWER HOLE					
ID of CF Shaft Hole	X			Y		
LH	MECHANIC	QA	SPV	MECHANIC	QA	SPV
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						

3	OUTSIDE DIAMETER OF CAM ROLLER					
OD CAM ROLLER	X			Y		
RH	MECHANIC	QA	SPV	MECHANIC	QA	SPV
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						

CAM FOLLOWER AND PUSH ROD MEASUREMENT CHECK SHEET

WO NUMBER		PART NAME	CAM FOLLOWER & PUSHROD	DOCUMENT NO.	RC/QR03/EG/10/01
ENGINE MODEL	12V140-1,3 Series	PART NUMBER		PUBL. DATE	10/08/2008
ENGINE S/N		START DATE/TIME		REVISION NO.	04
MACHINE MODEL		FINISH DATE/TIME		PAGE	

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3	OUTSIDE DIAMETER OF CAM ROLLER					
OD CAM ROLLER	X			Y		
LH	MECHANIC	QA	SPV	MECHANIC	QA	SPV
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						

BENDING OF PUSH ROD						
PUSH ROD	IN			EX		
RH	MECHANIC	QA	SPV	MECHANIC	QA	SPV
1						
2						
3						
4						
5						
6						

PUSH ROD	IN			EX		
LH	MECHANIC	QA	SPV	MECHANIC	QA	SPV
1						
2						
3						
4						
5						
6						

CAM FOLLOWER AND PUSH ROD MEASUREMENT CHECK SHEET

WO NUMBER		PART NAME	CAM FOLLOWER & PUSHROD	DOCUMENT NO.	RC/QR03/EG/10/01
ENGINE MODEL	12V140-1,3 Series	PART NUMBER		PUBL. DATE	10/08/2008
ENGINE S/N		START DATE/TIME		REVISION NO.	04
MACHINE MODEL		FINISH DATE/TIME		PAGE	

5 of 5

MECHANIC INSPECTION COMMENT :

INSPECTED BY :

INSPECTED & APPROVED BY :

MECHANIC (MECH)

QUALITY ASSURANCE (QA)

PROD. SUPERVISOR (SPV)



PT SAPTAINDRA SEJATI

**ACCESSORY DRIVE MEASUREMENT &
ASSEMBLY**

CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

HD785-7

RO NO :

ENGINE SECTION



PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

ACCESSORY DRIVE ASSEMBLY CHECK SHEET

KOMATSU ENGINE 12V140-1,-3

WO NUMBER		PART NAME	ACCESSORY DRIVE	NO	RC/QA/EG/062
ENGINE MODEL	12V140-1,-3	PART NUMBER		PUBLISHE	27/07/2009
ENGINE S/N		START DATE / TIME		REVISION	02
MACHINE MODEL		FINISH DATE / TIME			

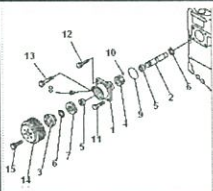
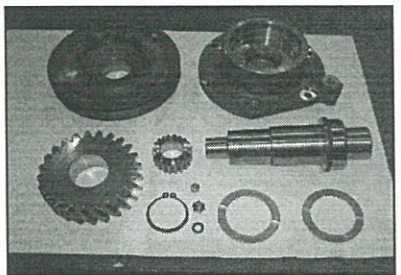
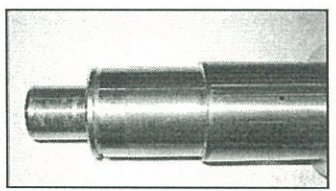
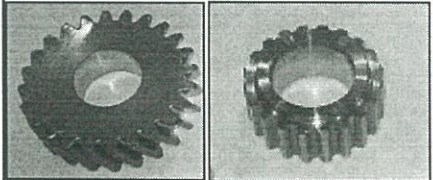
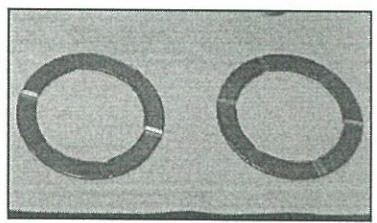
ASSEMBLY BY	
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NO	CHECK POINT	STD		REPAIR LIMIT	
1	Outside diameter of shaft	39.975 ~ 39.959		-	
	Inside diameter of bushing	40.000 ~ 40.025		-	
	Clearance between OD shaft & ID bushing	0.106 ~ 0.025		0,150	
2	Outside diameter of shaft	30.085 ~ 30.070		-	
	Inside diameter of gear	30.000 ~ 30.021		-	
	Interference between OD shaft & ID gear	0.085 ~ 0.049		0,048	
3	Outside diameter of shaft for Hub contact	35.950 ~ 35.968		-	
	Inside diameter of Hub (contact with shaft)	35,925		-	
	Interference between OD shaft & ID hub	0.025 ~ 0.043		-	
NO	CHECK POINT	ACTUAL		VISUAL CHECK (OK or NOT OK)	
		MECHANIC	QA/GL		
1	Outside diameter of Shaft	Front	X		
			Y		
		Rear	X		
			Y		
	Inside diameter of Bushing	Front	X		
			Y		
		Rear	X		
			Y		
	Clearance between outside diameter shaft and Inside diameter bushing	Front	Min		
			Max		
		Rear	Min		
			Max		
	Outside diameter of Shaft	X1			
		Y1			
		X2			
		Y2			
2	Inside diameter of Gear	X1			
		Y1			
		X2			
		Y2			
	Interference between out side diameter shaft and Inside diameter bushing	1	Min		
			Max		
		2	Min		
			Max		
3	Outside diameter of shaft for Hub contact	X1			
		Y1			
		X2			
		Y2			
	Inside diameter of Hub (contact with shaft)	X1			
		Y1			
		X2			
		Y2			
	Interference between out side diameter shaft and Inside diameter Hub	1	Min		
			Max		
		2	Min		
			Max		

ACCESSORY DRIVE ASSEMBLY CHECK SHEET

KOMATSU ENGINE 12V140-1,-3

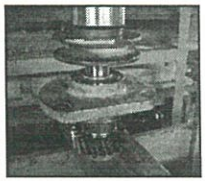
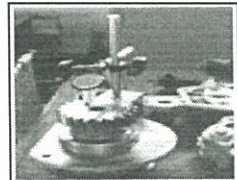
WO NUMBER		PART NAME		ACCESSORY DRIVE	NO	RC/QA/EG/062
ENGINE MODEL		PART NUMBER			PUBLISHE	27/07/2009
ENGINE S/N		START DATE / TIME			REVISION	02
MACHINE MODEL		FINISH DATE / TIME				

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN		REMARKS (SKETCH, PHOTOS, COMMENT.)																																																																																				
					MEC	QA/GL																																																																																					
	All Parts		Confirm that Parts which have order do not damage, scratch and corrosion.				 <table border="1" style="font-size: 8px; margin-top: 5px;"> <thead> <tr> <th>Item</th> <th>Part No.</th> <th>Description</th> <th>Qty</th> </tr> </thead> <tbody> <tr><td>1</td><td>6215-21-3400</td><td>ACCESSORY DRIVE SHAFT</td><td>1</td></tr> <tr><td>2</td><td>6215-21-3410</td><td>HOUSING</td><td>1</td></tr> <tr><td>3</td><td>6215-21-3421</td><td>SHAFT</td><td>1</td></tr> <tr><td>4</td><td>6215-21-3430</td><td>HUB</td><td>1</td></tr> <tr><td>5</td><td>6215-21-3441</td><td>GEAR</td><td>1</td></tr> <tr><td>6</td><td>6127-21-6221</td><td>BUSHING</td><td>2</td></tr> <tr><td>7</td><td>6127-21-6261</td><td>METAL THRUST</td><td>2</td></tr> <tr><td>8</td><td>6215-21-3450</td><td>SEAL OIL</td><td>1</td></tr> <tr><td>9</td><td>6215-21-3470</td><td>SEAL OIL</td><td>1</td></tr> <tr><td>10</td><td>07802-21-1800</td><td>SEAL</td><td>1</td></tr> <tr><td>11</td><td>07843-70100</td><td>PLUG</td><td>1</td></tr> <tr><td>12</td><td>07800-72100</td><td>O RING (R-2)</td><td>1</td></tr> <tr><td>13</td><td>07800-62100</td><td>O RING (R-2)</td><td>1</td></tr> <tr><td>14</td><td>07800-72016</td><td>O RING (R-2)</td><td>1</td></tr> <tr><td>15</td><td>07800-62016</td><td>O RING (R-2)</td><td>1</td></tr> <tr><td>16</td><td>01810-81230</td><td>BOLT</td><td>2</td></tr> <tr><td>17</td><td>01810-81250</td><td>BOLT</td><td>1</td></tr> <tr><td>18</td><td>01811-81250</td><td>BOLT</td><td>1</td></tr> <tr><td>19</td><td>0287-61-3060</td><td>PULLEY HARDENING</td><td>1</td></tr> <tr><td>20</td><td>01810-80016</td><td>BOLT</td><td>6</td></tr> </tbody> </table>	Item	Part No.	Description	Qty	1	6215-21-3400	ACCESSORY DRIVE SHAFT	1	2	6215-21-3410	HOUSING	1	3	6215-21-3421	SHAFT	1	4	6215-21-3430	HUB	1	5	6215-21-3441	GEAR	1	6	6127-21-6221	BUSHING	2	7	6127-21-6261	METAL THRUST	2	8	6215-21-3450	SEAL OIL	1	9	6215-21-3470	SEAL OIL	1	10	07802-21-1800	SEAL	1	11	07843-70100	PLUG	1	12	07800-72100	O RING (R-2)	1	13	07800-62100	O RING (R-2)	1	14	07800-72016	O RING (R-2)	1	15	07800-62016	O RING (R-2)	1	16	01810-81230	BOLT	2	17	01810-81250	BOLT	1	18	01811-81250	BOLT	1	19	0287-61-3060	PULLEY HARDENING	1	20	01810-80016	BOLT	6
Item	Part No.	Description	Qty																																																																																								
1	6215-21-3400	ACCESSORY DRIVE SHAFT	1																																																																																								
2	6215-21-3410	HOUSING	1																																																																																								
3	6215-21-3421	SHAFT	1																																																																																								
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12	07800-72100	O RING (R-2)	1																																																																																								
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14	07800-72016	O RING (R-2)	1																																																																																								
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16	01810-81230	BOLT	2																																																																																								
17	01810-81250	BOLT	1																																																																																								
18	01811-81250	BOLT	1																																																																																								
19	0287-61-3060	PULLEY HARDENING	1																																																																																								
20	01810-80016	BOLT	6																																																																																								
1	Housing		Confirm that housing free from dirt, crack, corrosion and damage. Confirm that contact bearing free from corrosion and scratch. Confirm that oil hole clean and is not clogged. Confirm that bolt seat is not damaged. Set seal seat and O-ring position																																																																																								
2	SHAFT		Confirm that shaft is free from dirt, corrosion, crack, scratch and fitting. Confirm that contact seal is not scratch and worn.																																																																																								
3	GEAR		Confirm that gear is free from dirt, corrosion and crack. Confirm that gear is not fitting and damage. Confirm that gear in contact shaft is not wear, scratch and fitting.																																																																																								
4	Assembly Bushing on Housing		Install bushing in housing by using Bushing guide tools Confirm that bushing isn't damage and scratch. Confirm that hole in bushing has been in right position aligned with housing oil hole.																																																																																								
5	Assembly Shaft on Housing		Install seal by using guide tools Install trust metal 1 pcs in shaft which will contact with housing and lubricated by oil. Install Shaft in housing and lubricated by oil. Install Metal 1 pcs in shaft that will contact with gear. Install gear in shaft by using Hydraulic jack. Confirm that gear isn't scratch.																																																																																								

ACCESSORY DRIVE ASSEMBLY CHECK SHEET

KOMATSU ENGINE 12V140-1,-3

WO NUMBER				PART NAME	ACCESSORY DRIVE	NO	RC/QA/EG/062
ENGINE MODEL		12V140-1,-3		PART NUMBER		PUBLISHE	27/07/2009
ENGINE S/N				START DATE / TIME		REVISION	02
MACHINE MODEL				FINISH DATE / TIME			

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN		REMARKS (SKETCH, PHOTOS, COMMENTS)
					MEC	QA/GL	
6	Assembly Gear		Install gear in housing that has been assembly by using tools and press.				 
7	Finishing		Check end play Test rotation of drive. Actual : mm	Std 0.1-0.4 mm			

MECHANIC ASSEMBLY COMMENT :

ASSEMBLY BY

INSPECTED & APPROVED BY

MECHANIC (MECH)

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION



PT SAPTAINDRA SEJATI

OIL ADAPTER ASSEMBLY

CHECKSHEET

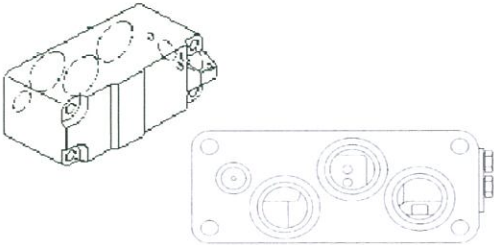
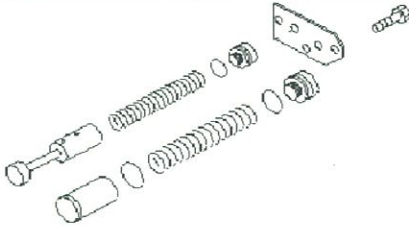
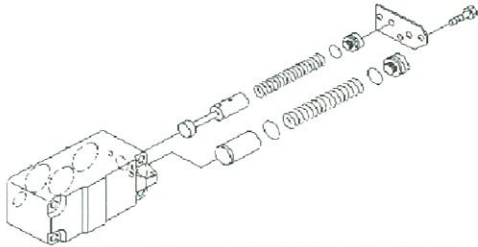
KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

HD785-7

RO NO :

ENGINE SECTION

ASSEMBLY CHECK SHEET		WO NUMBER		DOCUMENT NO		RC/QR04/EG/13/01	
OIL ADAPTER		COMP.MODEL		12V140E-1		PUBLIC DATE	
SAA12V140-3		COMP.S/N				REVISION NO	
		UNIT MODEL				PAGE	
PROCESS		START		ASSEMBLY BY			
		FINISH					
NO	PART TO INSTALL OR FIT	CONDITION TOBE ASSAMBLY	SPECIFICATION	SIGNATURE		SKETCH DRAWING OR PHOTHOES	
				MECH	GL/QA		
1	BODY	⚠ IMPORTANT ⚠ - Oil adapter body must be free from crack,dent and rust. - All thread and holes are in good condition and clean.					
2	INNER PART	- Valve must be free from abnormal worn ,scratch,dent ,pitching,and corotion. - Spring must be free from corotion,worn and broken. - Plate must be free from worn ,crack and broken.					
3	INSTALL REGULATOR VALVE ,BYPASS VALVE TO HOUSING	- Coat oil in hole of body ,by pass valve and regulator valve - Install regulator and by pass valve. - Install shim 1,0mm to oil cooler bypass valve.					
4	INSTALL SPRING,PLUG AND PLATE	- Coat spring and plug with oil engine - Install spring to valve. - Install o ring to plug - install plate with tool hidraulic press. ⚠ IMPORTANT ⚠ must be carefull and safety at install plate.check plug and spring,at insert to body.					
5	MOUNTING PLATE	Install plate and tighten mounting plate Actual: kgm	3 Kgm				

NOTE :

ASSEMBLY BY	INSPECTED BY	APPROVED BY
MECHANIC	Quality Assurance(QA)	GL/SPV PRODUCTION



PT SAPTAINDRA SEJATI

WATERPUMP ASSEMBLY CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

HD785-7

RO NO :

ENGINE SECTION

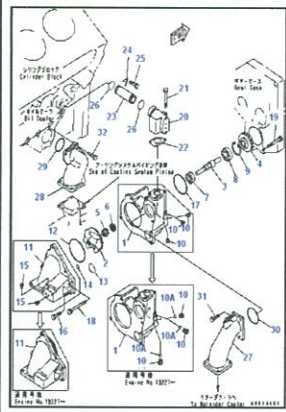
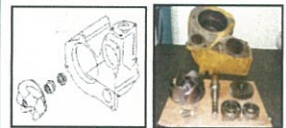
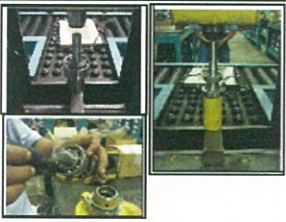
ASSEMBLY CHECK SHEET

WATER PUMP

6D170&12V140 SERIES

WO NUMBER		PART NAME	WATER PUMP	DOCUMENT NO.	RC/QA/EG/028
ENGINE MODEL		PART NUMBER		PUBLIC DATE	25/03/2010
ENGINE S/N		START DATE / TIME		REVISION NO.	3
MACHINE MODEL		FINISH DATE / TIME		PAGE	

ASSEMBLY BY		All dimension in millimetres
		All torques in Kilogram metres

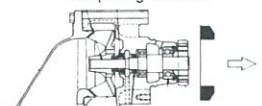
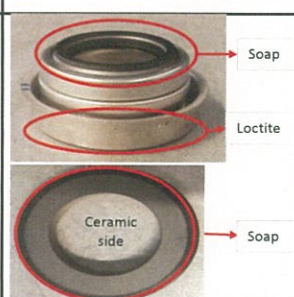
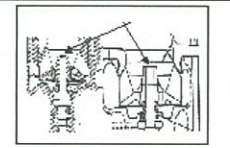
NO.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN		REMARKS (SKETCH, PHOTOS, COMMENTS)
					MEC	QA/GL	
1	Pump Body		Pump body must be free from crack, dent and rush Clean All Parts throughly and make sure that all burns have been remove. Always check the contact of the following parts before assembly. See checking contact between parts. Marking with paint maker, for all of bolts and nuts after tightening by torque wrench	See maintenance standard for standard dimension			
2	Housing		Housing must be free from crack, dents and rust All threads are in good condition and clean Make sure that face (clearance area) is in good and clean				
3	Other Parts		Recheck condition of bearings, oil seal, shaft, impeller, water seal, gear, snap ring, bolts and washers should be free from crack, rust, dents, scratch and pitting	Refer and follow PSN			There are many PSN for the Water pump 
4	Insertion of Oil Seal		Oil seal seat must be clean and dry (free from oil, fuel or water) Coat the out side of oil seal with LG 1215. Insert oil seal using special tools Coat the seal lip with "grease" sufficiently	Use oil seal installer LG 1215 G2-LI			
5	Insertion Bearing to Shaft		Insert small bearing and big bearing to shaft using special tools Make sure that bearings center to shaft when assembly For 12V140 Series Before inserting the small bearing and the big bearing, make sure that the shaft is new shaft with model P/n.6219-61-1230 (shaft end stamping reading "S5")	Use special Tools and Press by Hyd.Jack			 Make sure that bearings center to the when assembly

ASSEMBLY CHECK SHEET

WATER PUMP

6D170&12V140 SERIES

WO NUMBER		PART NAME	WATER PUMP	DOCUMENT NO.	RC/QA/EG/028
ENGINE MODEL		PART NUMBER		PUBLIC DATE	25/03/2010
ENGINE S/N		START DATE / TIME		REVISION NO.	3
MACHINE MODEL		FINISH DATE / TIME		PAGE	

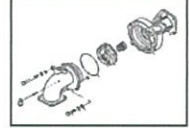
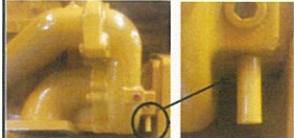
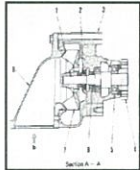
NO.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN		REMARKS (SKETCH, PHOTOS, COMMENTS)
					MEC	QA/GL	
6	Shaft Assembly		Coat the shaft (seal contact) by sufficient grease Bearings and bearing seat must be clean Insert shaft ass'y to the body and make sure that shaft ass'y is centering to body CAUTION Recheck and make sure that oil seal isn't hit or damage when inserting the shaft Check shaft from smooth rotation.	Use special Tools (Shaft Installer)			Snap Ring Position  Big Snap ring (at water pump) has face, please insert chamfer face to outside.  
7	Insertion of Drive Gear		Shaft and gear contact must be clean Coat the drive gear hole with engine oil Install Spacer. Insert drive gear to the shaft until contact to inner race bearing or stopper Standard of interference between Gear and Shaft Act. Interference : mm	Drive Gear Installer			When inserting the drive gear Please arrest the shaft at the End of the shaft  
8	Insertion of Water Seal		Make sure that the condition of water seal is free from scratch, crack, pitting, dents, corrosive, worn, and etc. Here is critical point, be careful Coat the outer side of water seal with adhesive Insert water seal to housing with special tool. Coat Ceramic surface of water seal with "toilet soap"	Follow and refer to the PSN LOCTITE 609 (green loctite) TOILET SOAP			 Soap Loctite Ceramic side Soap
9	Insertion of Impeller		Insert water seal using special tools Insert the seal ring to the groove at impeller and make sure that it is fit in the groove Coat the hole of the impeller with small quantity of Engine Oil be careful when insert the Impeller For 6D170 Series When inserting the impeller, adjust impeller from end of shaft using special tool (spacer) to assure Standard of Spacer Thickness Act. Spacer thickness = mm Clearance between impeller and connector / housing Standard Clearance Act. Clearance = mm For 12V140 Series Before inserting the impeller, make sure that impeller is new impeller model with P/n.6219-61-1470 (change the hole for the shaft) Act. Clearance = mm Insert the impeller to the shaft until contact to the end of the shaft.	SAE 15W-40 1.2 ~ 1.7 mm 0.23 ~ 0.66 mm Standard Clearance : 0.25 ~ 1.02 mm			    

ASSEMBLY CHECK SHEET

WATER PUMP

6D170&12V140 SERIES

WO NUMBER		PART NAME	WATER PUMP	DOCUMENT NO.	RC/QA/EG/028
ENGINE MODEL		PART NUMBER		PUBLIC DATE	25/03/2010
ENGINE S/N		START DATE / TIME		REVISION NO.	3
MACHINE MODEL		FINISH DATE / TIME		PAGE	

NO.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN		REMARKS (SKETCH, PHOTOS, COMMENTS)
					MEC	QA/GL	
10	Insertion of Housing		O-ring groove must be free from crack, scratch, pitting, dents and rust Check O-ring from scratch Insert O-ring and check from twist and out of it's groove Insert Housing/connection and tighten mounting bolts carefully and properly Tightening torque : M8 = 3.2 + 0.3 kgm Act. Torque M8 = Kgm M10 = 6.7 + 0.7 kgm Act. Torque M10 = Kgm M12 = 11.5 + 1.0 kgm Act. Torque M12 = Kgm				   
	Hole Indicator		Install extension tube to hole indicator				
11	General Inspection		√ : Check - Check Shaft for smooth rotation OK - Recheck position of snap ring OK - Check loose bolts from mistake OK - Check all parts from improper installation OK - Check and keep completion parts OK				 1. Pump shaft 2. Water seal 3. Pump body 4. Water pump drive gear (No. of teeth: 25) 5. Ball bearing 6. Oil seal 7. Impeller 8. Pump cover a. From thermostat b. From radiator (water inlet) c. To oil cooler (water outlet) 
12	End play water pump		Measure End play of water pump shaft before delivery to Main line	Standard End play 0			

MECHANIC ASSEMBLY COMMENT

ASSEMBLY BY

INSPECTED & APPROVED BY

MECHANIC (MECH)

QUALITY ASSURANCE (QA)

PROD. SPV/GL



PT SAPTAINDRA SEJATI

COMONRAIL RH ASSEMBLY CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

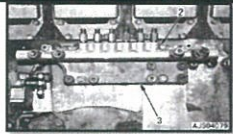
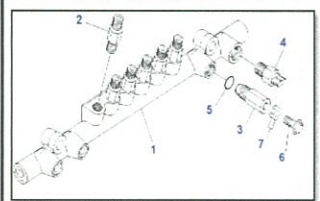
HD785-7

RO NO :

ENGINE SECTION

CHECK SHEET COMMON RAIL ASSEMBLY RH KOMATSU ENGINE 12V140-3 SERIES	WO NUMBER		DOCUMENT NO	QR00-SA-008-01
	ENGINE MODEL	12V140-3 SERIES	PUBLICATION DATE	8-Oct-13
	ENGINE S/N		REVISION NO	00
	MACHINE MODEL		PAGE	1 of 1

PROCESS	START DATE / TIME		ASSEMBLY BY	
	FINISH DATE / TIME			

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN			REMARKS (SKETCH, PHOTOS, COMMENTS)
					MEC	QA	SPV	
1	Common Rail Element	Housing	- Inspect Housing for cracks by visual check. No Crack - Before assembling ,wash & clean the Housing.	√= CHECK				
2	ALL PARTS	Flow Damper	- Make sure all Flow Damper are ready to used (free Crack ; worn; corrosion and dirt) - Before assembling , wash and clean the parts. (Allow them to dry by air blow) - Specification of spare parts are correct	√= CHECK				
		Plate,Plunger, Spring and Ball	-Make sure All inner Parts are ready to used (free Crack ; worn; corrosion and dirt) - Before assembling , wash and clean the parts. (Allow them to dry by air blow) - Specification of spare parts are correct	√= CHECK				
3	PRESSURE LIMITER	Fuel Pressure Limiter and Plug	- Make sure all parts are ready to used (free Crack ; worn; corrosion and dirt) Make sure Thread of Hexagon Plug front & rear and Thread of Pressure Limiter good condition(not broken)	√= CHECK				
4	Assembly Common Rail		- Coat inside surface of Housing and Inner parts with engine oil (SAE # 30) -Install Plate to Main Housing -Install Plunger to spring then Install to Flow Damper. and Tighten Flow damper (2): Actual : kgm -Install Ball & Hexagon Plug on front & rear of Housing and tighten Plug use L key socket: - Before tighten Plug to the Housing coat a thin of adhesive LT-2 to the thread of Plug(High streng type). Actual : kgm -Install Fuel pressure Limiter (3) with Picco Ring (5) to Main housing and Tighten. Actual : kgm -Install Fuel pressure sensor (4). to Main housing and Tighten. Actual : kgm	13 ± 1.0 kgm (Reman Manual) 5 ± 0.5 kgm (Komatsu KES Plug Standard) 9 ± 0.5 kgm (Reman Manual) 5 Kgm (Reman Manual)				 

MECHANIC ASSEMBLY COMMENT :

ASSEMBLY BY	INSPECTION & APPROVED BY
MECHANIC (MECH)	QUALITY ASSURANCE (QA)
	GL/SPV PRODUCTION



PT SAPTAINDRA SEJATI

COMONRAIL LH ASSEMBLY CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

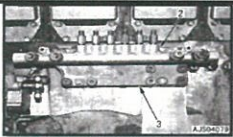
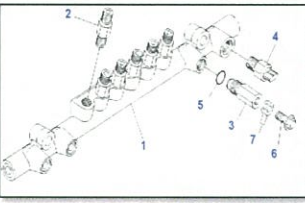
HD785-7

RO NO :

ENGINE SECTION

CHECK SHEET COMMON RAIL ASSEMBLY LH KOMATSU ENGINE 12V140-3 SERIES	WO NUMBER		DOCUMENT NO	QR00-SA-008-01
	ENGINE MODEL	12V140-3 SERIES	PUBLICATION DATE	8-Oct-13
	ENGINE S/N		REVISION NO	00
	MACHINE MODEL		PAGE	1 of 1

PROCESS	START DATE / TIME		ASSEMBLY BY	
	FINISH DATE / TIME			

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN			REMARKS (SKETCH, PHOTOS, COMMENTS)
					MEC	QA	SPV	
1	Common Rail Element	Housing	- Inspect Housing for cracks by visual check. No Crack - Before assembling ,wash & clean the Housing.	√= CHECK				
2	ALL PARTS	Flow Damper	- Make sure all Flow Damper are ready to used (free Crack ; worn; corrosion and dirt)	√= CHECK				
		Plate,Plunger, Spring and Ball	-Make sure All inner Parts are ready to used (free Crack ; worn; corrosion and dirt) - Before assembling , wash and clean the parts. (Allow them to dry by air blow) - Specification of spare parts are correct	√= CHECK				
3	PRESSURE LIMITER	Fuel Pressure Limiter and Plug	- Make sure all parts are ready to used (free Crack ; worn; corrosion and dirt) Make sure Thread of Hexagon Plug front & rear and Thread of Pressure Limiter good condition(not broken)	√= CHECK				
4	Assembly Common Rail		- Coat Inside surface of Housing and Inner parts with engine oil (SAE # 30) -Install Plate to Main Housing -Install Plunger to spring then Install to Flow Damper. and Tighten Flow damper (2): Actual : kgm -Install Ball & Hexagon Plug on front & rear of Housing and tighten Plug use L key socket: - Before tighten Plug to the Housing coat a thin of adhesive LT-2 to the thread of Plug(High streng type). Actual : kgm -Install Fuel pressure Limiter (3) with Picco Ring (5) to Main housing and Tighten. Actual : kgm -Install Fuel pressure sensor (4). to Main housing and Tighten. Actual : kgm	13 ± 1.0 kgm (Reman Manual) 5 ± 0.5 kgm (Komatsu KES Plug Standard) 9 ± 0.5 kgm (Reman Manual) 5 Kg (Reman Manual)				 

MECHANIC ASSEMBLY COMMENT :

ASSEMBLY BY

INSPECTION & APPROVED BY

MECHANIC (MECH)

QUALITY ASSURANCE (QA)

PROD. SUPERVISOR (SPV)



PT SAPTAINDRA SEJATI

OIL COOLER ASSEMBLY CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

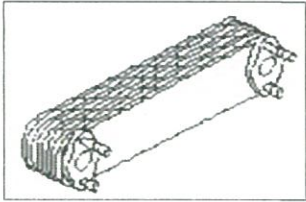
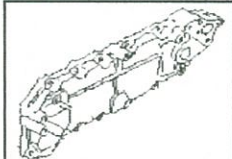
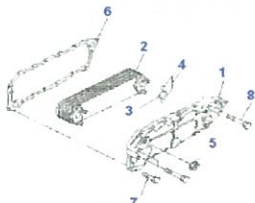
HD785-7

RO NO :

ENGINE SECTION

ASSEMBLY CHECK SHEET OIL COOLER KOMATSU ENGINE 12V140-3	WO NUMBER		DOCUMENT NO	RC/QR04/EG/06/01
	ENGINE MODEL	SAA12V140E-3	PUBLICATION DATE	27/07/2010
	ENGINE S/N		REVISION NO	00
	MACHINE MODEL		PAGE	1 of 1

PROCESS	START DATE / TIME	ASSEMBLY BY
	FINISH DATE / TIME	

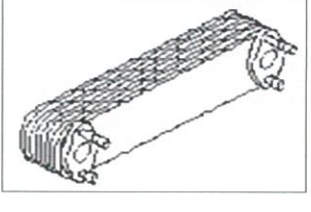
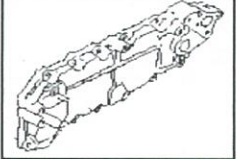
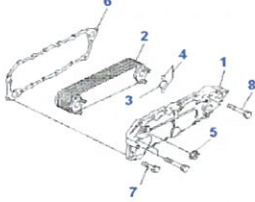
NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN			REMARKS (SKETCH, PHOTOS, COMMENTS)	
					MEC	QA	SPV		
1	Oil Cooler Element	Leakage	- Inspect for cracks and leakage by oil cooler flusher. Test and check leakage on core element. No Leakage	√= CHECK No Leakage					
		Mounting surface	- Inspect for corrosion, scratches and wear Depth of corrosion, scratch and wear No corrosion No scratch No wear	√= CHECK Depth ≤0.2mm					
		Bolt Hole	- Inspect tread portion appear corrosion, scratch and wear Depth of corrosion, scratch and wear No corrosion No scratch No wear	√= CHECK Depth ≤0.4mm					
2	Oil Cooler Housing	Mounting surface	- Inspect for corrosion, scratches and wear Depth of corrosion, scratch and wear No corrosion No scratch No wear	√= CHECK Depth ≤0.2mm					
		Bolt Hole	- Inspect tread portion appear corrosion, scratch and wear Depth of corrosion, scratch and wear No corrosion No scratch No wear	√= CHECK Depth ≤0.4mm					
3	Assembly element Oil Cooler		- Fit O ring or gasket on housing - Install washer and nut then tighten. - Tighten Torque : Actual : Kgm	Apply LG-6 M10 = 2.8~ 3.8 Kgm M12 = 5.6~ 6.6 Kgm (Komatsu assembly manual)					

COMMENT :

ASSEMBLY BY	INSPECTION & APPROVED BY
MECHANIC (MECH)	QUALITY ASSURANCE (QA)
	GL/SPV PRODUCTION

ASSEMBLY CHECK SHEET OIL COOLER KOMATSU ENGINE 12V140-3	WO NUMBER		DOCUMENT NO	RC/QR04/EG/06/01
	ENGINE MODEL	SAA12V140E-3	PUBLICATION DATE	27/07/2010
	ENGINE S/N		REVISION NO	00
	MACHINE MODEL		PAGE	1 of 1

PROCESS	START DATE / TIME		ASSEMBLY BY	
	FINISH DATE / TIME			

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN			REMARKS (SKETCH, PHOTOS, COMMENTS)
					MEC	QA	SPV	
1	Oil Cooler Element	Leakage	- Inspect for cracks and leakage by oil cooler flusher. Test and check leakage on core element. No Leakage	√= CHECK No Leakage				 
		Mounting surface	- Inspect for corrosion, scratches and wear Depth of corrosion, scratch and wear No corrosion No scratch No wear	√= CHECK Depth ≤0.2mm				
		Bolt Hole	- Inspect tread portion appear corrosion, scratch and wear Depth of corrosion, scratch and wear No corrosion No scratch No wear	√= CHECK Depth ≤0.4mm				
2	Oil Cooler Housing	Mounting surface	- Inspect for corrosion, scratches and wear Depth of corrosion, scratch and wear No corrosion No scratch No wear	√= CHECK Depth ≤0.2mm				 
		Bolt Hole	- Inspect tread portion appear corrosion, scratch and wear Depth of corrosion, scratch and wear No corrosion No scratch No wear	√= CHECK Depth ≤0.4mm				
3	Assembly element Oil Cooler		- Fit O ring or gasket on housing - Install washer and nut then tighten. - Tighten Torque : Actual : Kgm	Apply LG-6 M10 = 2.8~ 3.8 Kgm M12 = 5.6~ 6.6 Kgm (Komatsu assembly manual)				

COMMENT :

ASSEMBLY BY	INSPECTION & APPROVED BY
MECHANIC (MECH)	QUALITY ASSURANCE (QA)
	GL/SPV PRODUCTION



PT SAPTAINDRA SEJATI

OIL COOLER

WATERLEAK TESTER

CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

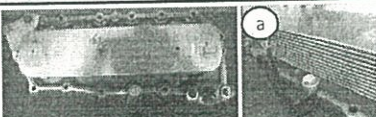
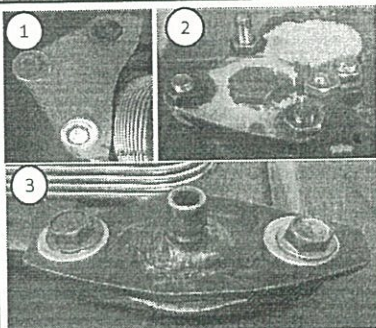
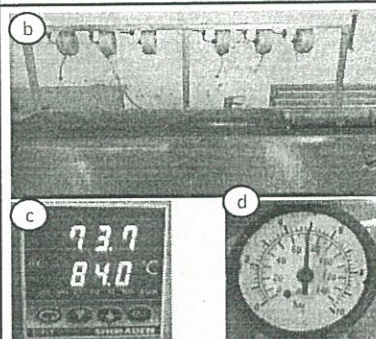
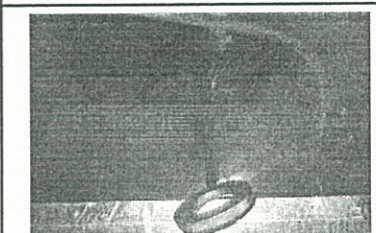
HD785-7

RO NO :

ENGINE SECTION

WATER LEAK TEST CHECK SHEET OIL COOLER ASSY KOMATSU ENGINE 12V140 SERIES	WO NUMBER		Eks UNIT CODE	
	ENGINE MODEL	12V140 SERIES	PUBLISHED DATE	22-Jan-15
	ENGINE S/N		REVISION NO	00
	MACHINE MODEL		DOCUMENT NO	

PROCESS	START	TEST BY
	FINISH	

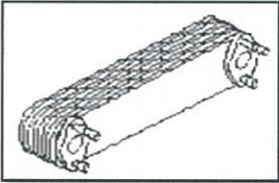
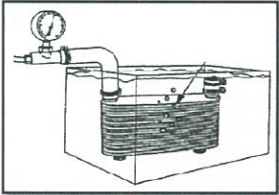
NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN			REMARKS (SKETCH, PHOTOS, COMMENTS)
					MEC	QA	SPV	
1	Core Element		- Check Element normal condition. (broken from bend,rust etc) - Install hook/hanger (a).	Visual check				
2	Plug,adapter and Air inlet connector		- Install plug on housing (1). - Install adapter and use O ring (2) - Install inlet air connector and use sealant or rubber for this connector (3)					
3	Water leak tester and indicator		- Prepare water leak tester tool (b) - Setting water temperature at standar range ' (c). Actual : °C - Setting air pressure on Regulator valve (d) Actual : Kg/cm	Water leak tester tool STD range tmpratur 75° ~ 90°C (Shop manual) 3 ~ 4 kg/cm (Reman manual)				
4	Water leak test oil cooler		- Install coupler and test oil cooler assy - Time for test Actual : Minutes	5 Minutes (Reman manual)				

Comment :

ASSEMBLY BY	INSPECTION & APPROVED BY
MECHANIC (MEC)	QUALITY ASSURANCE (QA)
	PROD. SUPERVISOR (SPV)

AIRLEAK TEST CHECK SHEET OIL COOLER ELEMENT KOMATSU ENGINE 12V140 SERIES	WO NUMBER		DOCUMENT NO	QR00-SA-008-01
	ENGINE MODEL	12V140 SERIES	PUBLICATION DATE	
	ENGINE S/N		REVISION NO	01
	MACHINE MODEL		PAGE	1 of 1

PROCESS	START DATE / TIME		TESTED BY	
	FINISH DATE / TIME			

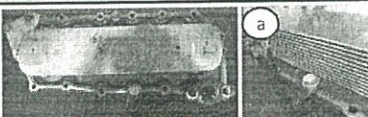
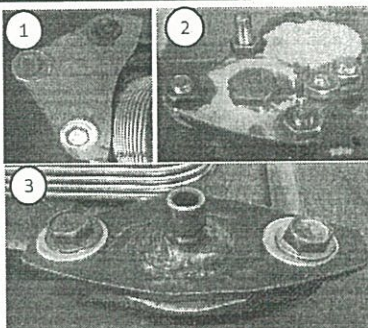
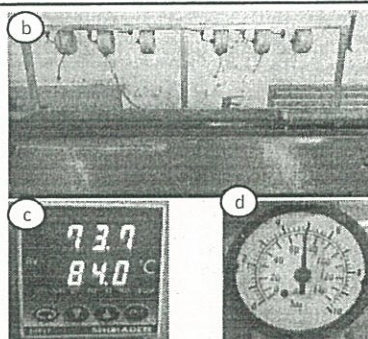
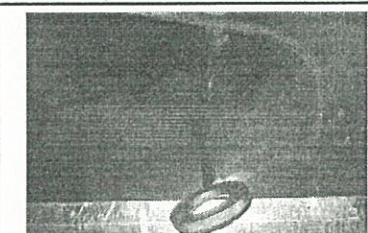
NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN			REMARKS (SKETCH, PHOTOES, COMMENTS)
					MEC	QA	SPV	
1	Oil Cooler Element	Air Leak test	- Inspect Element for cracks by visual check. No Crack	√ = CHECK				 
2	Test pressure		Check for Crack by Air pressure test for Element a). Air pressure: Actual : Kgm/cm2 b). Water Temperature Actual : °C c). Time for test Actual : Minutes No Leakage	3 ~ 4 kg/cm2 (Reman Manual) 70°C (Reman Manual) 5 minutes (Reman Manual) √ = CHECK No Leakage				 

COMMENT :

TESTED BY	INSPECTION & APPROVED BY	
MECHANIC (MECH)	QUALITY ASSURANCE (QA)	GL/SPV PRODUCTION

WATER LEAK TEST CHECK SHEET OIL COOLER ASSY KOMATSU ENGINE 12V140 SERIES	WO NUMBER		Eks UNIT CODE	
	ENGINE MODEL	12V140 SERIES	PUBLISHED DATE	22-Jan-15
	ENGINE S/N		REVISION NO	00
	MACHINE MODEL		DOCUMENT NO	

PROCESS	START	TEST BY
	FINISH	

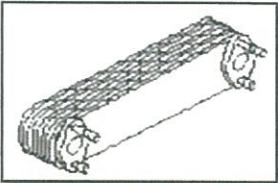
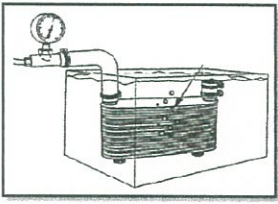
NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN			REMARKS (SKETCH, PHOTOS, COMMENTS)
					MEC	QA	SPV	
1	Core Element		- Check Element normal condition. (broken from bend, rust etc) - Install hook/hanger (a).	Visual check				
2	Plug, adapter and Air inlet connector		- Install plug on housing (1). - Install adapter and use O ring (2) - Install inlet air connector and use sealant or rubber for this connector (3)					
3	Water leak tester and indicator		- Prepare water leak tester tool (b) - Setting water temperature at standar range '(c). Actual : °C - Setting air pressure on Regulator valve (d) Actual : Kg/cm	Water leak tester tool STD range tmpratur 75° ~ 90°C (Shop manual) 3 ~ 4 kg/cm (Reman manual)				
4	Water leak test oil cooler		- Install coupler and test oil cooler assy - Time for test Actual : Minutes	5 Minutes (Reman manual)				

Comment :

ASSEMBLY BY	INSPECTION & APPROVED BY
MECHANIC (MEC)	QUALITY ASSURANCE (QA)
	PROD. SUPERVISOR (SPV)

AIRLEAK TEST CHECK SHEET OIL COOLER ELEMENT KOMATSU ENGINE 12V140 SERIES	WO NUMBER		DOCUMENT NO	QR00-SA-008-01
	ENGINE MODEL	12V140 SERIES	PUBLICATION DATE	
	ENGINE S/N		REVISION NO	01
	MACHINE MODEL		PAGE	1 of 1

PROCESS	START DATE / TIME		TESTED BY	
	FINISH DATE / TIME			

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN			REMARKS (SKETCH, PHOTOS, COMMENTS)
					MEC	QA	SPV	
1	Oil Cooler Element	Air Leak test	- Inspect Element for cracks by visual check. No Crack	√ = CHECK				 
2	Test pressure		Check for Crack by Air pressure test for Element a). Air pressure: Actual : Kgm/cm2 b). Water Temperature Actual : °C c). Time for test Actual : Minutes No Leakage	3 ~ 4 kg/cm2 (Reman Manual) 70°C (Reman Manual) 5 minutes (Reman Manual) √ = CHECK No Leakage				 

COMMENT :

TESTED BY	INSPECTION & APPROVED BY	
MECHANIC (MECH)	QUALITY ASSURANCE (QA)	GL/SPV PRODUCTION



PT SAPTAINDRA SEJATI

TURBOCHARGE ASSEMBLY

CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

HD785-7

RO NO :

ENGINE SECTION

ASSEMBLY CHECK SHEET TURBOCHARGER ENGINE ALL SERIES	WO NO.		PART NAME		DOCUMENT NO.	RC/QR04/EG/08/01
	ENGINE MODEL		PART NUMBER		PUBLIC DATE	12/08/2010
	ENGINE S/N		START DATE /TIME		REVISION NO.	02
	MACHINE MODEL		FINISH DATE / TIME		PAGE	

No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN			NOTES, DIAGRAMS COMMENTS
					MEC	QA	GL	
			Tightening torque :					
			Act. KTR 100 = kgm	KTR 100	4.25 ± 0.25			
			Act. KTR 110 = kgm	KTR 110	4.5 ± 0.25			
			Act. KTR 130 = kgm	KTR 130	8.5 ± 0.5			
			Act. KTR 150 = kgm	KTR 150	9.5 ± 0.5			
			Check End Play					
			Actual KTR 100 : = mm	KTR 100	Standard Repair limit	0.06 to 0.11 0,18		
			Actual KTR 110 : = mm	KTR 110	Standard Repair limit	0.08 to 0.13 0,18		
			Actual KTR 130 : = mm	KTR 130	Standard Repair limit	0.08 to 0.13 0,18		
			Actual KTR 150 : = mm	KTR 150	Standard Repair limit	0.08 to 0.13 0,18		
			Check Radial Play					
			Actual KTR 100 : = mm	KTR 100	Standard Repair limit	0.20 to 0.44 0,6		
10	Housing	Turbine Housing	Take care not to damage the blower impeller and turbine impeller when installing the blower housing and turbine housing					 
			For KTR 110 , KTR 150 Install the Turbine housing into the center housing and the rotor assembly. Apply a thin coat of anti seiz compound to the flange section of turbine housing. Install lock plate, tighten mounting bolt. Apply a thin coat of anti seiz compound to the thread of bolt.					
			Tighten mounting bolt :					
			Act. KTR 110 = kgm	KTR 110	5.0 ± 0.5			
			Act. KTR 150 = kgm	KTR150	5.75 ± 0.25			
								 
		Blower Impeller housing	Install blower housing, aligning it with the center housing and the rotor assembly. Apply a thin coat of Grease to the flange section.					
			For KTR 100 , KTR 130 Install the blower housing and turbine housing into the center housing and the rotor assembly.					
			Install V-Band clamp and tighten then torque of the V-Band locknut.					
			Act. KTR 100 = kgm	KTR 100	0.9 ± 0.1			
			Act. KTR 130 = kgm	KTR 130	0.9 ± 0.1			

ASSEMBLY CHECK SHEET TURBOCHARGER ENGINE ALL SERIES	WO NO.		PART NAME		DOCUMENT NO.	RC/QR04/EG/08/01
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No.	PARTS	CHECK POINT	CONDITIONS TO BE CHECKED	SPEC.	SIGN			NOTES, DIAGRAMS COMMENTS
					MEC	QA	GL	
			Install V-Band clamp and tighten then torque of the V-Band locknut.					  
			Act. KTR 110 = kgm	KTR 110	0.9 ± 0.1			
			Act. KTR 150 = kgm	KTR150	0.9 ± 0.1			
			Measure the clearance in the radial direction in the following manner.					
			Depress the end of the rotor in the radial direction with a finger.					
			Measure the clearance between the blower impeller and the housing by using filler gauge.					
			Tolerance (Min.) :					
			Act. KTR 100 = mm	KTR 100	0,15			
			Act. KTR 110 = mm	KTR 110	0,2			
			Act. KTR 130 = mm	KTR 130	0,2			
			Act. KTR 150 = mm	KTR 150	0,2			
AFTER COMPLETING THE ASSEMBLY, ROTATE THE ROTOR ASSEMBLY WHILE DEPRESSING IT LIGHTLY AND MAKE SURE THAT THE ROTOR CAN BE ROTATED SMOOTHLY WITHOUT BINDING.								

[illegible]

ASSEMBLY BY	INSPECTION & APPROVAL BY :	
_____ MECHANIC (MEC)	_____ QUALITY ASSURANCE (QA)	_____ GL/SPV PRODUCTION



PT SAPTAINDRA SEJATI

TURBOCHARGE ASSEMBLY

CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

HD785-7

RO NO :

ENGINE SECTION



PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

ASSEMBLY CHECK SHEET TURBOCHARGER ENGINE ALL SERIES	WO NO.		PART NAME		DOCUMENT NO.	RC/QR04/EG/08/01
	ENGINE MODEL		PART NUMBER		PUBLIC DATE	12/08/2010
	ENGINE S/N		START DATE / TIME		REVISION NO.	02
	MACHINE MODEL		FINISH DATE / TIME		PAGE	

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			Act. KTR 100 = kgm	KTR 100	4.25 ± 0.25			
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			Act. KTR 150 = kgm	KTR 150	9.5 ± 0.5			
			Check End Play					
			Actual KTR 100 : = mm	KTR 100	Standard Repair limit	0.06 to 0.11 0,18		
			Actual KTR 110 : = mm	KTR 110	Standard Repair limit	0.08 to 0.13 0,18		
			Actual KTR 130 : = mm	KTR 130	Standard Repair limit	0.08 to 0.13 0,18		
			Actual KTR 150 : = mm	KTR 150	Standard Repair limit	0.08 to 0.13 0,18		
			Check Radial Play					
			Actual KTR 100 : = mm	KTR 100	Standard Repair limit	0.20 to 0.44 0,6		
			Actual KTR 110 : = mm	KTR 110	Standard Repair limit	0.25 to 0.44 0,6		
			Actual KTR 130 : = mm	KTR 130	Standard Repair limit	0.25 to 0.43 0,6		
			Actual KTR 150 : = mm	KTR 150	Standard Repair limit	0.25 to 0.43 0,6		
10	Housing	Turbine Housing	Take care not to damage the blower impeller and turbine impeller when installing the blower housing and turbine housing					 
			For KTR 110 , KTR 150 Install the Turbine housing into the center housing and the rotor assembly. Apply a thin coat of anti seiz compound to the flange section of turbine housing. Install lock plate, tighten mounting bolt. Apply a thin coat of anti seiz compound to the thread of bolt.					
			Tighten mounting bolt :					
			Act. KTR 110 = kgm	KTR 110	5.0 ± 0.5			
			Act. KTR 150 = kgm	KTR150	5.75 ± 0.25			
	Blower Impeller housing		Install blower housing, aligning it with the center housing and the rotor assembly. Apply a thin coat of Grease to the flange section.					 
			For KTR 100 , KTR 130 Install the blower housing and turbine housing into the center housing and the rotor assembly.					
			Install V-Band clamp and tighten then torque of the V-Band locknut.					
			Act. KTR 100 = kgm	KTR 100	0.9 ± 0.1			
			Act. KTR 130 = kgm	KTR 130	0.9 ± 0.1			

ASSEMBLY CHECK SHEET TURBOCHARGER ENGINE ALL SERIES	WO NO.		PART NAME		DOCUMENT NO.	RC/QR04/EG/08/01
	ENGINE MODEL		PART NUMBER		PUBLIC DATE	12/08/2010
	ENGINE S/N		START DATE /TIME		REVISION NO.	02
	MACHINE MODEL		FINISH DATE / TIME		PAGE	

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					MEC	QA	GL																		
			Install V-Band clamp and tighten then torque of the V-Band locknut. <table border="1"> <tr> <td>Act. KTR 110 =</td> <td>kgm</td> <td>KTR 110</td> <td>0.9 ± 0.1</td> </tr> <tr> <td>Act. KTR 150 =</td> <td>kgm</td> <td>KTR150</td> <td>0.9 ± 0.1</td> </tr> </table>	Act. KTR 110 =	kgm	KTR 110	0.9 ± 0.1	Act. KTR 150 =	kgm	KTR150	0.9 ± 0.1														
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[illegible]

MECHANIC (MEC)	QUALITY ASSURANCE (QA)	GL/SPV PRODUCTION
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REV.02



PLANT REBUILD CENTER

TENSION PULLEY

ASSEMBLY CHECKSHEET

KOMATSU ENGINE SAA12V140E-3

MACHINE MODEL

HD 785-7

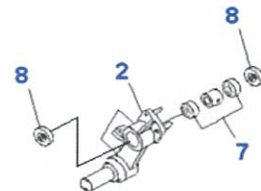
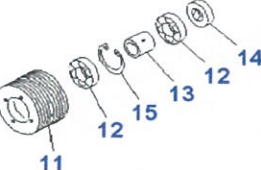
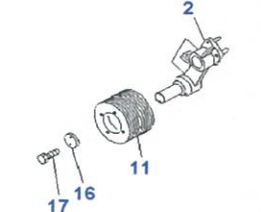
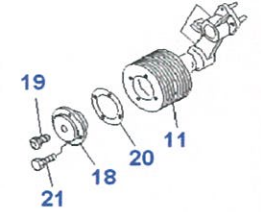
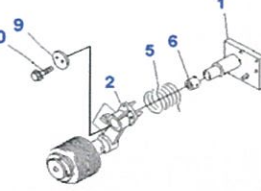
WO NO:

SUB ASSY SECTION

ASSEMBLY CHECK SHEET TENSION PULLEY KOMATSU ENGINE 12V140E-3	WO NUMBER		DOCUMENT NO	RC/QR04/EG/10/01
	ENGINE MODEL		PUBLISHED DATE	27/08/2010
	ENGINE S/N		REVISION NO	00
	MACHINE MODEL		PAGE	

1 of 2

PROCESS	START DATE / TIME		ASSEMBLY BY	
	FINISH DATE / TIME			

NO.	PARTS	ASSEMBLY PROCESS	SPEC	SIGN			REMARKS (SKETCH, PHOTOS, COMMENTS)
				MEC	QA	SPV	
1	All parts	- All parts must be free from dust, rust and dent - All parts have been cleaning - No crack, no scratch, no wear - Always check the contact of the following parts before assembly.					
2	Tension Lever	- Make sure tension lever is clean and free from damage. - Reamer bearing seat hole, then install outer race and needle bearing (7) - *Using press tool to install bearing. - Install dust, seal (8) before assembly with shaft					
3	Pulley ass'y	- Make sure tension lever is clean and free from damage. - Install snap ring (15) on groove in the pulley - Install bearing (12) and spacer (13) in the pulley - *Using press tool to install bearing. - Install Oil seal (14) before assembly with lever					
4	Assembly Pulley and Lever	- Assembly pulley ass'y (11) and tension lever (2) - Install cover lock (16), then tighten mounting bolts (17). Actual : kgm	3.2 ± 0.3 kgm				
5	Pulley Cover and Relief Valve	- Fit gasket (20), then install cover (18) - Install mounting bolts (21), then tighten Actual : kgm - Install relief valve (19) then tighten	6.7 ± 0.7 kgm				
6	Assembly Lever assy and tension shaft	- Install Inner race (5) to the shaft (1) - Install spring (5), then assembly lever ass'y (2) - Install plate (9), install mounting bolts (10) then tighten. Actual : kgm	6.7 ± 0.7 kgm				

ASSEMBLY CHECK SHEET TENSION PULLEY KOMATSU ENGINE 12V140E-3	WO NUMBER		DOCUMENT NO	RC/QR04/EG/10/01
	ENGINE MODEL		PUBLISHED DATE	27/08/2010
	ENGINE S/N		REVISION NO	00
	MACHINE MODEL		PAGE	

NO.	PARTS	CONDITIONS TO BE CHECKED	SPEC	SIGN			REMARKS (SKETCH, PHOTOS, COMMENTS)
				MEC	QA	SPV	
7	Fitting grease and plug	- Install fitting grease and plug, then tighten - Give all of surface pulley with grease to protect from rust					

Notes :

ASSEMBLY BY MECHANIC (MEC)	INSPECTED & APPROVED BY QUALITY ASSURANCE (QA)	GL/SPV PRODUCTION
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PT SAPTAINDRA SEJATI

FUEL SUPPLY PUMP DRIVE ASSEMBLY CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

HD785-7

RO NO :

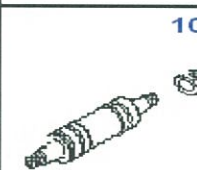
ENGINE SECTION

FUEL PUMP DRIVE ASSEMBLY CHECK SHEET KOMATSU ENGINE 12V140-3

HD785-7					
PC2000-8					

WO NUMBER		PART NAME	FUEL PUMP DRIVE	DOCUMENT NO	RC/QR04/EG/07/01
ENGINE MODEL	12V140-3	PART NUMBER		PUBLIC DATE	01/08/2009
ENGINE SERIAL NO.		START DATE / TIME		REVISION NO.	00
MACHINE MODEL		FINISH DATE / TIME		PAGE	

ASSEMBLY BY		All dimension in millimetres 1 of 3
		All Torques in Kilogram metres

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC	SIGN		REMARKS (SKETCH, PHOTOS, COMMENT)																																																																																																																																																																																																																																																																																																																																																																																																																				
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1	ALL PARTS		- Make sure all parts are ready to used (free Crack ; worn; corrosion and dirt) - Specification of spare parts are correct - Before assembling , wash and clean the parts. - Allow them to dry by air blow				 <table border="1"><thead><tr><th>ITEM NO.</th><th>DESCRIPTION</th><th>QTY</th><th>REMARKS</th></tr></thead><tbody><tr><td>1</td><td>GEAR 10</td><td>1</td><td>1000000</td></tr><tr><td>2</td><td>GEAR 29</td><td>1</td><td>1000000</td></tr><tr><td>3</td><td>GEAR 10</td><td>1</td><td>1000000</td></tr><tr><td>4</td><td>GEAR 29</td><td>1</td><td>1000000</td></tr><tr><td>5</td><td>GEAR 10</td><td>1</td><td>1000000</td></tr><tr><td>6</td><td>GEAR 29</td><td>1</td><td>1000000</td></tr><tr><td>7</td><td>GEAR 10</td><td>1</td><td>1000000</td></tr><tr><td>8</td><td>GEAR 29</td><td>1</td><td>1000000</td></tr><tr><td>9</td><td>GEAR 10</td><td>1</td><td>1000000</td></tr><tr><td>10</td><td>GEAR 29</td><td>1</td><td>1000000</td></tr><tr><td>11</td><td>GEAR 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2	GEAR		- Install gear then tighten lock nut (10 and 29) Actual <table><tr><td>RH</td><td>Kgm</td></tr></table> - Turn the shaft to make sure that condition is normal - Turn the case so that face "A" of drive case facing up up set on hydraulic press stand and press fit the bushing (Front side) with special tool. - Align the hole lubrication of bushing to the hole lubrication at the drive case. - Make sure the position of hole lubrication was corrected - Measure inside diameter of the bushing after press fit to the drive case Standard Inside Diameter of Bushing : Actual <table><thead><tr><th>SIDE</th><th></th><th>FRONT BUSHING</th><th>REAR BUSHING</th></tr></thead><tbody><tr><td rowspan="2">X-X</td><td>LH</td><td></td><td></td></tr><tr><td>RH</td><td></td><td></td></tr><tr><td rowspan="2">Y-Y</td><td>LH</td><td></td><td></td></tr><tr><td>RH</td><td></td><td></td></tr></tbody></table> - Apply a thin coat of engine oil to the bushing and the shaft - Trial install the shaft to the bushing by hand rotate the shaft and make sure rotation of shaft as smooth	RH	Kgm	SIDE		FRONT BUSHING	REAR BUSHING	X-X	LH			RH			Y-Y	LH			RH			STANDARD TORQUE 19 ~ 20 Kgm 53.972 ~ 54.069 mm		10     																																																																																																																																																																																																																																																																																																																																																																																																	
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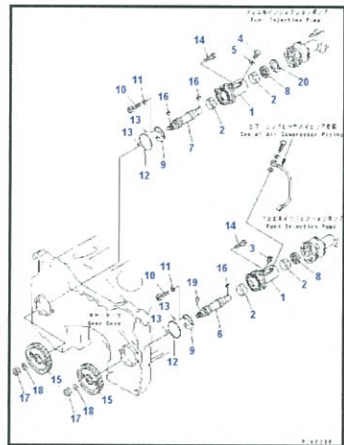
FUEL PUMP DRIVE ASSEMBLY CHECK SHEET KOMATSU ENGINE 12V140-3

HD785-7					
PC2000-8					

WO NUMBER		PART NAME	FUEL PUMP DRIVE	DOCUMENT NO	RC/QR04/EG/07/01
ENGINE MODEL	12V140-3	PART NUMBER		PUBLIC DATE	01/08/2009
ENGINE SERIAL NO.		START DATE / TIME		REVISION NO.	00
MACHINE MODEL		FINISH DATE / TIME		PAGE	

2 of 3

2 of 3

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC	SIGN		REMARKS (SKETCH, PHOTOS, COMMENT)																				
					MEC	GL/QA																					
3	SHAFT		- Make sure the shaft that contact bushing area free from scratch - Make sure key groove and thread is good condition - Measure outside diameter of the shaft Standard Outside Diameter of Shaft : Actual Outside diameter of shaft <table border="1"> <thead> <tr> <th colspan="2" rowspan="2">SIDE</th><th colspan="2">SHAFT</th></tr> <tr> <th>FRONT</th><th>REAR</th></tr> </thead> <tbody> <tr> <td rowspan="2">X-X</td><td>LH</td><td></td><td></td></tr> <tr> <td>RH</td><td></td><td></td></tr> <tr> <td rowspan="2">Y-Y</td><td>LH</td><td></td><td></td></tr> <tr> <td>RH</td><td></td><td></td></tr> </tbody> </table> - If the shaft that contact surface with lip of the oil seal is worn can use sleeve for repair. - Install sleeve to the shaft and press fit with special tool - Set seat of the oil seal with sleeve - Before install the sleeve to the shaft coat inside surface of sleeve with adhesive LT-2 - After installing sleeve, confirm that there isn't scratch on the sleeve surface	SIDE		SHAFT		FRONT	REAR	X-X	LH			RH			Y-Y	LH			RH			53.934 ~ 53.984 mm			  
SIDE		SHAFT																									
		FRONT	REAR																								
X-X	LH																										
	RH																										
Y-Y	LH																										
	RH																										
4	Assembly Shaft on Drive Case		- Coat Inside surface of bushing and oil seal with engine oil (SAE # 30) - Install shaft in the drive case by hand from face A side • Dimension B of drive shaft : • Left bank = Short • Right bank = Long				   																				
5	Thrust Plate		- Set notch of thrust plate facing up install thrust plate to the thrust plate groove in the shaft and tighten bolts - Check that the shaft rotates smoothly - Before tighten bolt to the drive case coat a thin of adhesive LT-2 to the thread of bolt. Actual : kgm	3.15 ± 0.15 kgm			   																				

FUEL PUMP DRIVE ASSEMBLY CHECK SHEET

KOMATSU ENGINE 12V140-3

HD785-7					
PC2000-8					

WO NUMBER		PART NAME	FUEL PUMP DRIVE	DOCUMENT NO	RC/QR04/EG/07/01
ENGINE MODEL	12V140-3	PART NUMBER		PUBLIC DATE	01/08/2009
ENGINE SERIAL NO.		START DATE / TIME		REVISION NO.	00
MACHINE MODEL		FINISH DATE / TIME		PAGE	

3 of 3

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC	SIGN		REMARKS (SKETCH, PHOTOS, COMMENT)								
					MEC	GL/QA									
6	OIL SEAL		- Make sure that oil seal in good condition (oil seal PSN;P/N: 6215-71-3281) - Press fit oil seal in the drive case with special tool and then apply a thin coat of the engine oil to lip of the oil seal - Before install the oil seal to the housing coat outside surface of oil seal with liquid LG-7.												
7	KEY		- Knock key into key Groove - Make sure the key groove free from crack												
8	FINISHING		- Check end play <table border="1"><thead><tr><th colspan="3">Standard End play</th></tr></thead><tbody><tr><td rowspan="2">Act.</td><td>LH</td><td>mm</td></tr><tr><td>RH</td><td>mm</td></tr></tbody></table> - Turn the shaft to make sure that condition is normal	Standard End play			Act.	LH	mm	RH	mm	0.07 ~ 0.20 mm			
Standard End play															
Act.	LH	mm													
	RH	mm													

MECHANIC ASSEMBLY COMMENT

ASSEMBLY BY

INSPECTED & APPROVED BY



PT SAPTAINDRA SEJATI

OIL FILTER HEAD ASSEMBLY

CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

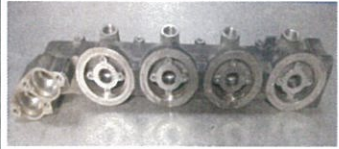
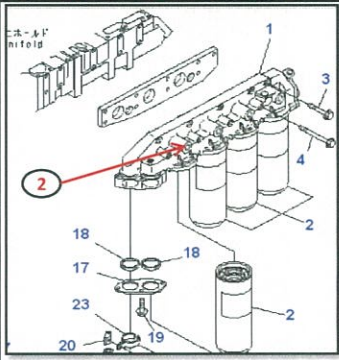
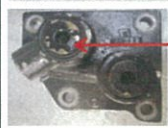
HD785-7

RO NO :

ENGINE SECTION

CHECK SHEET OIL FILTER HEAD ASSEMBLY KOMATSU ENGINE 12V140-3 SERIES	WO NUMBER		DOCUMENT NO	RC/QR04/EG/14/01
	ENGINE MODEL	12V140-3 SERIES	PUBLICATION DATE	12-Oct-13
	ENGINE S/N		REVISION NO	00
	MACHINE MODEL		PAGE	1 of 1

PROCESS	START DATE / TIME	ASSEMBLY BY	
	FINISH DATE / TIME		

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC.	SIGN			REMARKS (SKETCH, PHOTOS, COMMENTS)
					MEC	QA	SPV	
1	Head assy	Housing	- Inspect Main Housing for cracks by visual check. - Check O ring area contact(oil inlet and outlet)free from scratch and chipping. No Crack - Before assembling ,wash & clean the Housing.	√= CHECK				
2	Adapter & Cover Plate	Adapter Cover Plate	- Make sure all Adapter are ready to used (free Crack ; worn; corrosion ;dirt and check thread condition is OK). -Make sure Cover plate is ready to used (free Crack ; worn; corrosion; dirt and not bending) - Before assembling , wash and clean the parts. (Allow them to dry by air blow) - Specification of spare parts are correct	√= CHECK √= CHECK				   
3	Completion parts	Plug and Mounting Bolt	- Make sure all parts are ready to used (free Crack ; worn; corrosion and dirt) Make sure thread of plug condition are OK,thread of all Bolt are OK. - Before assembling ,wash & clean the Housing.	√= CHECK				   
4	Assembly Oil filter head		-Install Adapter to main housing and mounting the Adapter.tightening torque: Actual : kgm -Install plug no.2 with washer gasket to main housing,Quantity of Plug are 4 pcs.tightening torque: Actual : kgm Install cover plate to housing and mounting the cover. tightening torque: Actual : kgm Install Hexagon plug no.1 and torque this plug. tightening torque: Actual : kgm	3.1 ± 0.3 kgm (Komatsu drawing std) 9.5 ~ 12.5 kgm (Komatsu drawing std) 5.7 ± 0.7 kgm (Komatsu Torque std) 5.5 ~ 6.0 kgm (Komatsu drawing std)				 
5	Final Inspection		-Recheck all plug and bolt from loosen -Recheck marks					

MECHANIC ASSEMBLY COMMENT :

ASSEMBLY BY	INSPECTION & APPROVED BY



PT SAPTAINDRA SEJATI

FAN PULLEY DRIVE

CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

HD785-7

RO NO :

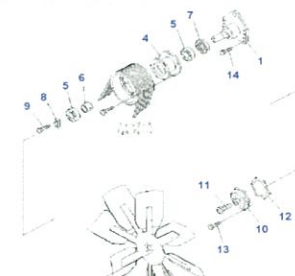
ENGINE SECTION

FAN PULLEY ASSEMBLY CHECK SHEET

KOMATSU ENGINE 12V140-3

WO NUMBER		PART NAME	FAN PULLEY	DOCUMENT NO	RC/QR04/EG/11/01
ENGINE MODEL	12V140-3	PART NUMBER		PUBLIC DATE	23/09/2009
ENGINE SERIAL NO.		START DATE / TIME		REVISION NO.	00
MACHINE MODEL		FINISH DATE / TIME		PAGE	1 of 1

ASSEMBLY BY		All dimension in millimetres
		All Torques in Kilogram metres

NO	PARTS	CHECK POINT	CONDITION TO BE CHECKED	SPEC	SIGN		REMARKS (SKETCH, PHOTOS, COMMENT)																																																																																																				
					MEC	GL/SPV/QA																																																																																																					
1	ALL PARTS		- Make sure all parts are ready to used (free Crack ; worn; corrosion and dirt) - Specification of spare parts are correct - Before assembling , wash and clean the parts. - Allow them to dry by air blow				 <table><thead><tr><th>Item</th><th>Part No.</th><th>Part Name</th><th>Qty</th><th>Serial No.</th></tr></thead><tbody><tr><td>1</td><td>6215-61-4100</td><td>FAN PULLEY ASSY</td><td>1</td><td>10909</td></tr><tr><td>2</td><td>6215-61-3610</td><td>SHAFT</td><td>1</td><td>10909</td></tr><tr><td>3</td><td>07043-00011</td><td>PLUG</td><td>1</td><td>10909</td></tr><tr><td>4</td><td>07020-00000</td><td>FITTING GREASE</td><td>1</td><td>10909</td></tr><tr><td>5</td><td>6215-61-3326</td><td>CASE</td><td>1</td><td>10909</td></tr><tr><td>6</td><td>06200-00013</td><td>BEARING BALL</td><td>2</td><td>15306</td></tr><tr><td>7</td><td>06200-00013</td><td>BEARING BALL</td><td>2</td><td>10909 15305</td></tr><tr><td>8</td><td>6215-61-3326</td><td>SPACER</td><td>1</td><td>10909</td></tr><tr><td>9</td><td>07012-00010</td><td>SEAL OIL</td><td>1</td><td>10909</td></tr><tr><td>10</td><td>04000-00015</td><td>HOLDER</td><td>1</td><td>10909</td></tr><tr><td>11</td><td>01010-01435</td><td>BOLT</td><td>2</td><td>12733</td></tr><tr><td>12</td><td>01010-01435</td><td>BOLT</td><td>2</td><td>10909 12732</td></tr><tr><td>13</td><td>6215-61-3326</td><td>SPACER</td><td>1</td><td>10909</td></tr><tr><td>14</td><td>07020-00000</td><td>VALVE RELIEF</td><td>1</td><td>10909</td></tr><tr><td>15</td><td>01010-01275</td><td>GASKET</td><td>1</td><td>10909</td></tr><tr><td>16</td><td>01010-01275</td><td>BOLT</td><td>6</td><td>11463</td></tr><tr><td>17</td><td>01010-01275</td><td>BOLT</td><td>6</td><td>10909 11462</td></tr><tr><td>18</td><td>01010-01240</td><td>BOLT</td><td>8</td><td>11463</td></tr><tr><td>19</td><td>01010-01240</td><td>BOLT</td><td>8</td><td>10909 11462</td></tr></tbody></table>	Item	Part No.	Part Name	Qty	Serial No.	1	6215-61-4100	FAN PULLEY ASSY	1	10909	2	6215-61-3610	SHAFT	1	10909	3	07043-00011	PLUG	1	10909	4	07020-00000	FITTING GREASE	1	10909	5	6215-61-3326	CASE	1	10909	6	06200-00013	BEARING BALL	2	15306	7	06200-00013	BEARING BALL	2	10909 15305	8	6215-61-3326	SPACER	1	10909	9	07012-00010	SEAL OIL	1	10909	10	04000-00015	HOLDER	1	10909	11	01010-01435	BOLT	2	12733	12	01010-01435	BOLT	2	10909 12732	13	6215-61-3326	SPACER	1	10909	14	07020-00000	VALVE RELIEF	1	10909	15	01010-01275	GASKET	1	10909	16	01010-01275	BOLT	6	11463	17	01010-01275	BOLT	6	10909 11462	18	01010-01240	BOLT	8	11463	19	01010-01240	BOLT	8	10909 11462
Item	Part No.	Part Name	Qty	Serial No.																																																																																																							
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2	Case Housing		- Check all holes and then blow with air pressure - Install bearing and spacer to case housing with special tool . - Install oil seal on the back side - Install case housing ass'y to shaft and lock holder no 8. Tighten with mounting bolts Actual : kgm - After mounting between case housing and shaft fan pulley, do greasing until all surface bearing covered pass to nipple - Install spacer and pulley to case housing Tighten with mounting bolts Actual : kgm	18 ± 0.5 Kgm 11 - 12 Kgm			  																																																																																																				
3	Greasing pulley		- Give all of surface pulley with grease to protect from rust																																																																																																								

MECHANIC ASSEMBLY COMMENT

ASSEMBLY BY	INSPECTED & APPROVED BY
MECHANIC (MECH)	QUALITY ASSURANCE (QA)
	GL/SPV PRODUCTION



PT SAPTAINDRA SEJATI

CROSSHEAD SALVAGING PROCESS & INSPECTION CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

HD785-7

RO NO :

ENGINE SECTION

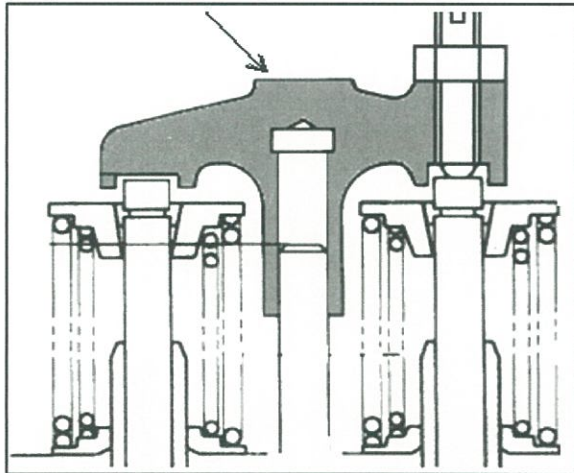


PT SAPTAINDRA SEJATI
PLANT REBUILD CENTER

SALVAGE CROSS HEAD PROCESS INSPECTION CHECKSHEET

WO NUMBER		PART NAME	CROSS HEAD	DOCUMENT NO.	QR06-SAL-006
ENGINE MODEL		PART NUMBER		PUBL. DATE	01/08/2008
ENGINE S/N		START DATE / TIME		REVISION NO.	01
MACHINE MODEL		FINISH DATE / TIME		PAGE	1 of 1

SKETCH DRAWING



Standard

Upper pad portion : Free from Wear

MECHANIC NAME

VISUAL CHECK	MECHANIC		QA	
	OK	NOT OK	OK	NOT OK
CRACK HAS BEEN CHECKED				
CHECK THREAD OF CROSSHEAD				
CHECK THREAD OF SCREW				
CHECK THREAD OF NUT				

MAN HOURS UTILIZATION

PROCESS	START	FINISH	MAN HOURS
Polishing			
Measuring			
TOTAL			

Physical Data

No.	Part Number	Part Name	Qty	New	Reuse
1		Cross head			
2		Screw			
3		Nut			

MECHANIC INSPECTION COMMENT :

INSPECTION BY

INSPECTION & APPROVED BY

MECHANIC

QUALITY ASSURANCE (QA)

GL/SPV PRODUCTION



PT SAPTAINDRA SEJATI

SUPPLY PUMP

ASSEMBLY

CHECKSHEET

KOMATSU ENGINE 12V140E-3 SERIES

MACHINE MODEL

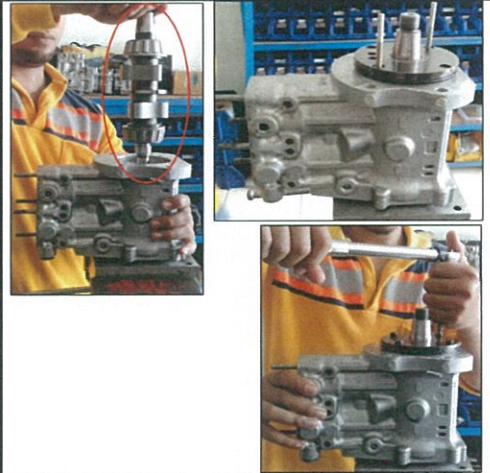
HD785-7

RO NO :

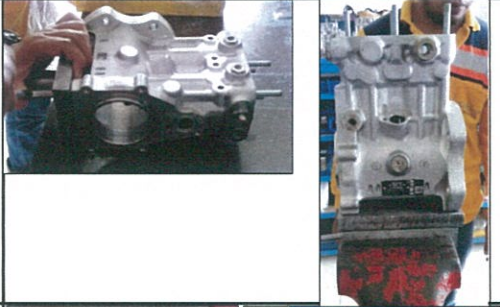
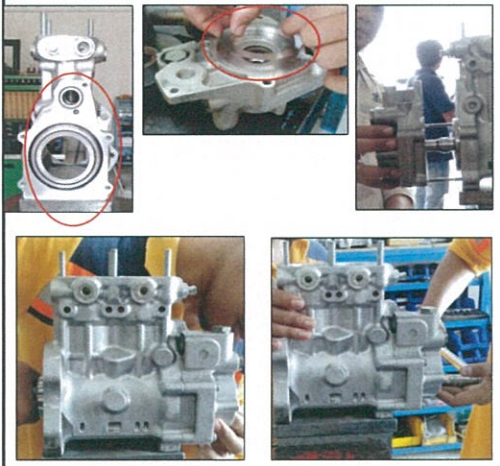
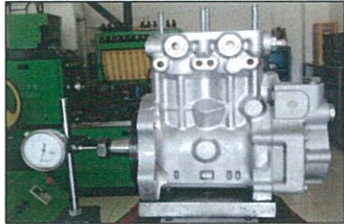
ENGINE SECTION

ASSEMBLY CHECK SHEET FUEL SUPPLY PUMP	WO NUMBER		DOCUMENT NO	
	COMP.MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	1
	UNIT MODEL		PAGE	

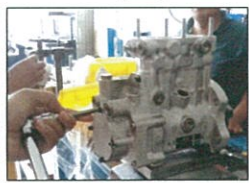
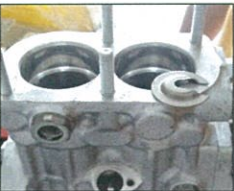
PROCESS	START		ASSEMBLY BY	
	FINISH			

NO	PART TO INSTALL OR FIT	CONDITION TO BE ASSEMBLY	SPECIFICATION	SIGNATURE		SKETCH DRAWING OR PHOTOS
				MECH	GL/QA	
	OUTER BEARING	1 Pastikan bidang contact antara outer bearing dan dudukan outer bearing tidak scratch, pitting dan bebas karat. 2 Siapkan special tool adapter install outer bearing. 3 posisikan outer bearing duduk di permukaan dudukan bearing di cover dan di housing. 4 Press outer bearing sampe tersentuh dengan stopper.				
	OIL SEAL	1 Pastikan oil seal dan dudukan oil seal tidak rusak, bersihkan dari kotoran dan debu pada bidang contact. 2 lapisi permukaan sisi luar oil seal dengan loctite. 3 Install oil seal menggunakan special tool. 4 Install oil seal pada cover bearing dan feed pump housing. 5 lapisi oli pada bibir oil seal.	LG 1207			
	RING & INNER BEARING CAMSHAFT	1 Pastikan ring dan inner bearing tidak rusak dan bebas dari kotoran dan karat. 2 pasang special tool ring spacer, 3 install ring menggunakan special tool .dengan metode press fit. 4 press ring sampai ring contact dengan tool spacer. 5 Install inner bearing menggunakan special tool sampai inner bearing contact dengan stopper.				
	CAMSHAFT TO HOUSING ASSY	1 Masukkan Camshaft assy ke housing supplypump. PENTING!! Saat memasukkan camshaft ,pastikan lip oil seal dilapisi oil dan tidak terlipat ataupun rusak. 2 Pasang stud dan pasang cover(2).dan pastikan oil seal tidak rusak dan terlipat. 3 Mounting baut cover dan torque bolt. TORQUE:	1,2 ~ 1,4 Kgm			

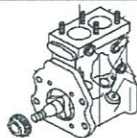
ASSEMBLY CHECK SHEET FUEL SUPPLY PUMP	WO NUMBER		DOCUMENT NO	
	COMP.MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	1
	UNIT MODEL		PAGE	

	HOUSING SUPPLY PUMP ASSY TO BRACKET	1 Install supply pump housing to bracket stand. 2 Torque bolt sesuai spesifikasi.	10 Kgm			
	HOUSING TRANSFER PUMP ASSY	1 Pastikan checkvalve yang akan di install tidak rusak,bebaskan dari kotoran dan debu. 2 Pastikan dudukan checkvalve tidak kotor,pitcing dan karat 3 Pasang checkvalve ,spring dan plug chamber,dan torque plug. TORQUE:	3 ~ 4 Kgm			
	FEED PUMP HOUSING TO MAIN HOUSING	1 Install oring dan sim 2 Install special tool 3 install feed pump housing. 4 pasang bolt dan torque. TORQUE:	STUD 1,2~1,4 Kgm			
	ENDPLAY CAMSHAFT	1 Ukur endplay camshaft Measured : <i>Note :Bila ukuran endplay tidak sesuai.maka adjust sim sampai mendapatkan ukuran endplay yang sesuai standart.</i>	0 ~ 0,06 mm			

ASSEMBLY CHECK SHEET FUEL SUPPLY PUMP	WO NUMBER		DOCUMENT NO	
	COMP.MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	1
	UNIT MODEL		PAGE	

FEED PUMP ASSEMBLY	1 Install side plate (inner)dengan lubang grove yang lebih besar menghadap ke bawah supply pump. 2 Install pump sub assembly. 3 Install Case (saat install case,posisikan tanda titik(.) di atas. 4 Install plate depan dan posisikan lubang grove yg lebih besar menghadap ke bawah. Mounting bolt plate dan torque bolt. TORQUE: 5 Install cover and mounting bolt . TORQUE:	0,3 ~ 0,4 Kgm		 1  2  3  4  5  6
PCV VALVE ASSEMBLY	1 Install tappet ke lubang PCV valve. 2 Install spring seat bagian bawah.install Piston shaft. 3 Install spring seat bagian atas. 4 Lock spring seat menggunakan special tool.			      

ASSEMBLY CHECK SHEET FUEL SUPPLY PUMP	WO NUMBER		DOCUMENT NO	
	COMP.MODEL		PUBLIC DATE	
	COMP.S/N		REVISION NO	1
	UNIT MODEL		PAGE	

PCV ELEMENT	1. Pastikan spring seat bagian atas spring sudah terkunci di housing supply pump. 2. Install oring ke PVC element. 3. lapisi Liquid gasket di area contact antara PVC element dan housing. 4. Install element dan torque nut. TORQUE: 5. Install side Plug and torque. TORQUE:	threebond 1215 2,5 ~ 3 Kgm 1,5 ~ 2 Kgm	  									
	1. Pasang special tool ke dalam lubang G sensor pada supply pump. 2. Geser rood special tool sampai berhenti pada titik puncak grove barrel sensor. 3. Kunci rood dan ukur panjang pergerakan rood dari titik nol. Actual measured 4. Ukur panjang G sensor dari titik fange sampai permukaan sensor. Actual measured 5. Saat menentukan SIM, pastikan jarak antara permukaan sensor dan barrel <table><tr><th>Part Number</th><th>Thickness</th></tr><tr><td>ND094096-0050</td><td>0,15</td></tr><tr><td>ND094096-0060</td><td>0,20</td></tr><tr><td>ND094096-0070</td><td>0,25</td></tr><tr><td>ND094096-0080</td><td>0,40</td></tr></table>	Part Number	Thickness	ND094096-0050	0,15	ND094096-0060	0,20	ND094096-0070	0,25	ND094096-0080	0,40	0,35 ~ 0,60 mm
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SUPPLYPUMP GEAR	1. Install Gear dan torque Nut. TORQUE:	18 ~ 20 Kgm										

ASSEMBLY BY

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MECHANIC

QUALITY ASSURANCE

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